

SLIGHTLY IMPROVED ZERO-FREE HALF-PLANES FOR THE QUASI-RIEMANN HYPOTHESIS

BAIYING LIU

ABSTRACT. Recently, a uniform $\frac{7}{8}$ zero-free half-plane for finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$ and its transfer to Dirichlet L -functions was given by OpenAI in [1]. In its proof, there are two chosen parameters $b = \frac{1}{8}, \ell = \frac{1}{6}$. In this note, via varying b and ℓ , following the same strategy, we slightly improve the bound $\frac{7}{8} = 0.875$ to be

$$B_r = \frac{34999}{40000} = 0.874975, \text{ (when } b_r = \frac{1}{8}, \ell_r = \frac{1}{6} + \frac{1}{10000}\text{);}$$

and

$$B_{\text{new}} = \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799\dots,$$

$$\text{(when } b_{\text{new}} = -\frac{4}{29} + \frac{230\sqrt{921}}{26709}, \ell_{\text{new}} = \frac{33 + 8\sqrt{921}}{1653}\text{)}.$$

This result has been formalized by Lean ([15]).

CONTENTS

Introduction	2
Part I. Summary of the $\frac{7}{8}$ proof	5
1. The source argument and its two-stage reduction	5
2. Arithmetic conventions	6
3. A criterion for zero-free continuation	8
4. Cubic theta series and the smoothed average	9
5. Zero detection by two character polynomials	14
6. The preliminary $11/12$ bound	15
7. The modified average and its direct estimate	16
8. Local cancellation and principal residues	17
9. The two moment estimates	19
10. Counting elements from simultaneous large values	21
11. Contour estimates and the endpoint inequality	23
12. Parameter choices and the final comparison	25
13. Mellin continuation and contradiction	25
14. Transfer to Dirichlet L -functions	26
Part II. A rational extension of the bound	26
15. Statement and choice of parameters	26
16. The direct estimate at balanced scales	27
17. Euler products and extraction of the principal term	29
18. The fixed moment parameter and the nonprincipal estimate	31
19. A rational polynomial inequality and the remaining norm ranges	33
20. Combining the estimates and the continuation contradiction	35

Part III. An algebraic extension of the bound	36
21. Algebraic parameters and the scope of the result	36
22. The direct estimate with free total length	37
23. Comparison of the character-count estimates	38
24. An exact polynomial proof of the endpoint	39
25. An algebraic lower bound for the exponent constraints	40
26. The auxiliary estimates and the final contradiction	41
Appendix A. Theta coefficient conventions	43
Appendix B. Index of cited results	43
References	44

INTRODUCTION

The quasi-Riemann hypothesis asks whether there is a fixed $B < 1$ such that $\zeta(s) \neq 0$ for $\operatorname{Re} s > B$, with its pole at $s = 1$ allowed. The uniform character version asks for the same B independently of conductor and height. A fixed half-plane supplies a uniform gap between zeros and the line $\operatorname{Re} s = 1$; classical zero-free regions instead narrow towards that line as conductor or height increases. For general analytic background, see [14, Chapter 5]; for the Dedekind zeta function of an abelian extension, and hence its Hecke L -function factors, a precise zero-free region allowing an exceptional real zero is [10, Theorem 3.1]. Zero-density estimates bound the number of zeros in a region; for example, see [11, Theorem 1.2].

We follow the 30 September 2026 preprint [1], retaining its numbering in all citations to it. The companion [12] gives a different 11/12 argument. OpenAI's public `math` repository has a formalization scope note [13, Scope] that lists the 7/8 statements for zeta, Dirichlet and finite-order Hecke functions, with principal poles excluded; the paper's later applications are outside that scope.

Let $F = \mathbb{Q}(\sqrt{-3})$. Let $\mathcal{Z}_F$ be the set of nontrivial zeros of all primitive finite-order Hecke L -functions over F , of arbitrary conductor; poles are excluded. Put

$$\beta_* := \sup(\{1/2\} \cup \{\operatorname{Re} \rho : \rho \in \mathcal{Z}_F\}).$$

We study the dependence of the cubic-theta argument on its length parameters. The stated 7/8 bound of [1] is the starting point for two deductions from its intermediate estimates. The stronger deduction is the following.

Main theorem. *The family supremum satisfies*

$$\beta_* \leq B_{\text{new}} := \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799\dots$$

Consequently every finite-order Hecke L -function over F and every Dirichlet L -function, including ζ , is zero-free for $\operatorname{Re} s > B_{\text{new}}$. Principal poles at $s = 1$ are allowed.

This is Theorem 21.1. The improvement is small:

$$\frac{7}{8} - B_{\text{new}} = 4.293020083\dots \times 10^{-5}.$$

The original 7/8 parameters therefore leave room for improvement. We expect B_{new} to be optimal for the particular direct and character-count estimates considered here, with $3/4 \leq \kappa \leq 1$, $1 \leq t \leq 3/2$, and positive scale exponents $l_x, l_y > \ell$, $b = l_y - l_x > 0$, $h = 1 - l_x + \ell > 0$. Proposition 25.1 proves a precise algebraic lower bound under three explicit exponent constraints; the broader optimality of the estimate system is stated as an expectation.

The choices of b and ℓ . In the original $7/8$ argument the lengths are chosen with $b = 1/8$ and $\ell = 1/6$. Here $b = l_y - l_x$ is the difference between the two summation exponents, and ℓ is the total exponent of the selected prime lengths. On balanced scales,

$$X = Z^{l_x}, \quad Y = Z^{l_y}, \quad l_x = \frac{1 - b - \ell}{2}, \quad l_y = \frac{1 + b - \ell}{2}.$$

The selected prime scales P_i satisfy $\prod_i P_i = Z^\ell$. For the admissible geometries used below and every $\varepsilon > 0$, the direct bound and the exponent of the principal Mellin term are respectively

$$|J_\eta(Z)| \ll_\eta Z^{l_x/2 + b/12 + \varepsilon}, \quad C_b(s) = s - \frac{4 + b}{6},$$

where J_η is the normalized cubic-theta average constructed in Part I and generalized in Part II. Matching these exponents gives

$$(0.1) \quad C_b(B) = \frac{l_x}{2} + \frac{b}{12} \iff B = B(\ell) := \frac{11}{12} - \frac{\ell}{4}.$$

Thus increasing ℓ lowers the comparison boundary. The allowable increase is constrained by the nonprincipal terms, whose exponents also depend on b . For the following parameter comparisons, write $E(b, \ell; \delta, x) = E_{B(\ell)}(h)$ for the relative nonprincipal exponent in (18.5), with $h = (1 + b + 3\ell)/2$ and the character-count exponent R_* of (18.2). Here $\delta = 2a - 1$ is the zero-class parameter, and $x = \bar{g}/\delta$ measures the normalized prime-sum size, as defined in Section 10; the polynomial comparison uses $(\delta, x) \in [0, 5/6] \times [0, 1/2]$. These comparisons use the fixed moment parameter $\kappa = 3/4$. The notation $\partial_\ell E$ denotes partial differentiation of $E(b, \ell; \delta, x)$ with respect to ℓ , holding b, δ, x fixed; the dependence of $B(\ell)$ and $h = (1 + b + 3\ell)/2$ on ℓ is included in this differentiation.

For the first improvement we keep the original length difference and increase the total prime length by $1/10000$:

$$b_r = \frac{1}{8}, \quad \ell_r = \frac{1}{6} + \frac{1}{10000}, \quad B_r = B(\ell_r) = \frac{7}{8} - \frac{1}{40000} = \frac{34999}{40000}.$$

The rational choice is deliberately conservative: b_r retains the original difference between the summation exponents, while $1/10000$ is a convenient perturbation rather than a distinguished or maximal value. Lemma 19.1 establishes the uniform bounds

$$E(1/8, 1/6; \delta, x) \leq -\frac{1}{5000}, \quad \partial_\ell E(1/8, \ell; \delta, x) \leq \frac{45}{32}.$$

Consequently, at $\ell = \ell_r$,

$$E(1/8, \ell_r; \delta, x) \leq -\frac{1}{5000} + \frac{45}{32 \cdot 10000} = -\frac{19}{320000} < 0.$$

Thus the improvement spends only part of the available error margin. Part II proves these bounds by an exact calculation in the Bernstein basis polynomials (defined in (19.3), in the proof of Lemma 19.1; see also [16, 17]) and verifies the remaining contour and length conditions.

For the second improvement we vary both parameters:

$$b_{\text{new}} = -\frac{4}{29} + \frac{230\sqrt{921}}{26709}, \quad \ell_{\text{new}} = \frac{33 + 8\sqrt{921}}{1653},$$

$$B_{\text{new}} = B(\ell_{\text{new}}) = \frac{1507 - 2\sqrt{921}}{1653}.$$

The algebraic values arise from the critical error class and a tangency condition. At $x = 1/2$, the count exponent is

$$R_*(\delta, 1/2) = \frac{185 - 288\delta + 96\delta^2}{185 - 138\delta}, \quad \partial_b E = \frac{R_*}{2} - \frac{1}{3}.$$

Hence a class with $R_* = 2/3$ imposes a constraint that cannot be relaxed by changing b . Its admissible parameter is the smaller root

$$288\delta^2 - 588\delta + 185 = 0, \quad \delta_c = \frac{49 - \sqrt{921}}{48}.$$

In particular, $\sqrt{921}$ comes from the discriminant $588^2 - 4 \cdot 288 \cdot 185 = 144 \cdot 921$, rather than from an arbitrary choice. At this class, $E = 0$ determines

$$\ell = \frac{5 - 6\delta_c}{9 + 18\delta_c} = \ell_{\text{new}},$$

and the stationary condition

$$\partial_\delta E(b, \ell_{\text{new}}; \delta_c, 1/2) = 0$$

determines $b = b_{\text{new}}$. These values balance the direct comparison with the largest non-principal exponent. The balancing equations locate the candidate; their solution alone does not establish the estimate for other classes. Part III derives the critical zero class, proves the endpoint inequality for every permitted zero and amplitude class by an exact quadratic identity, and verifies all auxiliary strict margins. The dual argument in Proposition 25.1 explains the algebraic obstruction suggested by these parameter choices. The notation new distinguishes this bound from the rational one. We remark that varying the moment parameter κ beyond the restricted range $3/4 \leq \kappa \leq 1$ used in that expectation may yield a further small improvement of B_{new} (by $\approx 4.4799 \times 10^{-8}$); we do not investigate this possibility here.

Both improvements retain the exact arithmetic formulas and principal residues of [1]. Their Euler correction is shown to be holomorphic below $7/8$. After the original argument establishes $\beta_* \leq 7/8$, the unweighted moment permits $\kappa = 3/4$, removing the loss associated with the original varying parameter. Strict comparison margins then give the same Mellin-continuation contradiction as in Part I. The analytic dependencies are indexed in Appendix B.

Here is the structure of the paper. Part I gives the detailed exposition of the $7/8$ proof and fixes all arithmetic and analytic notation, this is for convenience purpose. Part II establishes the shared changed-parameter estimates and gives the simpler rational extension

$$B_r = 34999/40000, \quad 7/8 - B_r = 1/40000.$$

Its calculation in the Bernstein basis polynomials is retained as a separate rational example; this bound is superseded by the stronger one in Part III. Part III varies both length parameters, proves the error-exponent inequality at the algebraic parameter choices, and discusses the resulting algebraic lower bound. Both extensions use the same continuation criterion and Dirichlet transfer from Part I.

Acknowledgments. OpenAI Codex (GPT 6.1 Sol Medium-Ultra) assisted with drafting, source checking, and the search for improved bounds. Anthropic Claude (Opus 5.5 Medium) assisted with auditing. The zero-free conclusions for B_r and B_{new} have Lean 4 formalizations for the finite-order Hecke family over F and its Dirichlet and zeta consequences [15]. The repository for this paper and its accompanying Lean files is [GitHub repository](#) [15].

Part I. Summary of the 7/8 proof

1. THE SOURCE ARGUMENT AND ITS TWO-STAGE REDUCTION

The argument compares two descriptions of one cubic-theta average: reflection bounds it directly, while Poisson summation isolates a principal Mellin integral involving $1/L_F(s, \eta)$. A zero makes two character polynomials simultaneously large; their moment bounds control the remaining Poisson terms. Part I makes those dependencies, the common-character hypotheses and the numerical exponent comparisons explicit. Its statements record the source argument using the proved results indexed in Appendix B. The accompanying explanations retain the decisive identities, exponent comparisons and contradiction, including calculations used in Parts II and III. Complete proofs of the imported results are given at the cited locations in [1]. Numbered statements have this paper's numbering; locators inside citations always refer to the cited work.

Theorem 1.1 (Main statement of [1, Theorem 1.1]). *Every finite-order Hecke L -function over $F = \mathbb{Q}(\sqrt{-3})$ has no zeros in*

$$(1.1) \quad \operatorname{Re} s > 7/8.$$

The same holds for every Dirichlet L -function, including ζ . Principal poles at $s = 1$ are allowed; the boundary $\operatorname{Re} s = 7/8$ is not included.

The Hecke and Dirichlet conclusions are proved below in Theorem 13.1 and Corollary 14.1. Here $L_F(s, \eta)$ denotes the Hecke L -function over F , and $\zeta(s)$ the Riemann zeta function; η denotes the character fixed in each application.

For the primitive Hecke family define

$$(1.2) \quad \beta_* = \sup \left(\{1/2\} \cup \left\{ \operatorname{Re} \rho : \begin{array}{l} L_F(\rho, \eta) = 0, \quad 1/2 \leq \operatorname{Re} \rho \leq 1, \\ \eta \text{ primitive and finite-order} \end{array} \right\} \right).$$

Poles are excluded. The inserted value $1/2$ gives $\beta_* \geq 1/2$; absolute convergence of the Euler product gives $\beta_* \leq 1$. The family in (1.2) comprises *all primitive finite-order Hecke characters over F , of arbitrary conductor*. Indeed, Poisson summation produces denominators $L_F^S(s, \eta \chi_{\bullet}(\overline{u}))$ in (4.7). The primitive inducing characters of these twists belong to that family, so the same β_* controls their zeros.

The two stages. Stage 1 proves $\beta_* \leq 11/12$. Stage 2 assumes, for contradiction, $\beta_* > 7/8$ and sets

$$(1.3) \quad \Delta = \beta_* - 7/8 \in (0, 1/24], \quad \kappa = 2\beta_* - 1 = 3/4 + 2\Delta \leq 5/6.$$

Group the nonprincipal terms by the location of the zeros of their associated twists; these groups are the zero classes of Section 5. In the second stage they satisfy

$$\delta = 2a - 1 \leq \kappa \leq \alpha := 5/6.$$

This restriction is essential in the source's character-count argument: the long-polynomial bound contains $(\alpha - \delta)r$, and its comparison with $r \leq t$ uses $\alpha - \delta \geq 0$. Thus the preliminary theorem cannot be omitted from this proof without replacing that input.

The parameter choice giving $7/8$. The normalized modified average has a direct bound $Z^{L+\varepsilon}$, while its principal residue has Mellin exponent $C(s)$, where

$$L = \frac{l_x}{2} + \frac{b}{12}, \quad b = l_y - l_x, \quad C(s) = s + \frac{l_x}{3} - \frac{5}{6} + \frac{\ell}{6}.$$

Under $l_x + l_y = 1 - \ell$, matching these exponents gives

$$C(B) = L \quad \Longleftrightarrow \quad B = \frac{11}{12} - \frac{\ell}{4}.$$

The reflected mean-square calculation in [1, Lemma 15.1, proof] contains the algebraic excess

$$\left(\frac{1 + 3\ell' - 2M'}{4}\right)_+ = \left(\frac{d_J + 5\ell - 1}{4}\right)_+, \quad \ell' = \ell - d_J, \quad M' = 1 - \ell - 2d_J.$$

For $0 \leq d_J \leq \ell$, this term vanishes for every subset exactly when $6\ell \leq 1$. The convenient choice $\ell = 1/6$ therefore gives $B = 7/8$ in the exponent matching above. With the paper's scales (7.1),

$$B = 7/8, \quad C(s) = s - 11/16, \quad C(7/8) = L = 3/16.$$

This exponent matching motivates the parameters; proving a strict saving for every non-principal contribution requires both large values from the zero detector and the two moment estimates. That saving continues every reciprocal $1/L_F(s, \eta)$ to the left of β_* , producing the contradiction.

Sections 2–4 introduce the characters, continuation criterion and theta average. Sections 5–6 give the detector and the 11/12 stage. Sections 7–12 develop the modified average, moments, counts and exponent savings. Sections 13–14 complete the continuation and Dirichlet transfer.

2. ARITHMETIC CONVENTIONS

Set

$$\omega = e^{2\pi i/3}, \quad \mathcal{O} = \mathbb{Z}[\omega], \quad \lambda = \sqrt{-3}, \quad q_a = |a|^2, \quad q_{\mathfrak{a}} = N\mathfrak{a}, \quad \varpi(a) = a/|a|.$$

$\mathcal{O}$ is norm-Euclidean. Ideals coprime to 3 have unique primary generators $a \equiv 1 \pmod{3}$. Ideal sums use these generators; sums over elements may also include units. For an integral ideal $\mathfrak{a}$, $N\mathfrak{a} = |\mathcal{O}/\mathfrak{a}|$. Write $v_p(a)$ for its prime valuation, $\text{rad}(a) = \prod_{p|a} p$, and μ for the ideal Möbius function. The notation sf in a sum means squarefree, and 1_E is the indicator of a condition E . Thus $(a, b) = 1$ means that the corresponding ideals are coprime. A prime outside S is called good.

Unless specified otherwise, $Z \geq 2$ is the asymptotic scale; $\varepsilon, \varepsilon_1 > 0$ are arbitrarily small losses. The symbols $\ll, \gg, \asymp$ refer to inequalities up to constants, and $x_+ = \max(x, 0)$. Auxiliary letters in separately stated lemmas have local scope: in particular, K in (4.4) is a norm bound, whereas K in Section 7 counts prime intervals. An annular weight is a smooth function supported on a fixed compact subinterval of $(0, \infty)$; its logarithmic derivatives are bounded. Constants A, J in height estimates denote finite nonnegative exponents or derivative orders fixed before the height cutoff.

Fix a finite set S containing primes above 6, primes in the full defining moduli of fixed character presentations, and primes of norm at most P_0 . Unless stated otherwise, ideal sums avoid S . The datum is fixed independently of the scale and the current summation variables; it may depend on an already fixed character η . Denote these fixed moduli, characters, ray group, exclusions and normalizations collectively by $\mathcal{A}$. The positive real P_0 is the fixed prime-norm cutoff.

Symbol	Meaning and scope
β_*, Δ, κ	Family zero supremum; $\Delta = \beta_* - 7/8$; $\kappa = 2\beta_* - 1$.
$\alpha = 5/6, \varpi(a)$	Fixed amplification slope; unit phase $a/ a $ (written $\alpha(a)$ in [1]).
ϵ_b, ϵ	Fixed zero-bin width (written e in [1]); arbitrarily small exponent loss.
$\mathfrak{a}(r)$	Amplified moment exponent $\max\{1, (1 + 5r)/6\}$ (written $e(r)$ in [1]).
$Z, U = Z^d, T_1$	Asymptotic scale; element norm scale; zero-detection height.
l_x, l_y, b, h, ℓ	Averaging exponents; $b = l_y - l_x$; combined frequency exponent; total auxiliary prime length.
z_{cap}, z_0	Prime-length cap in (9.2); real contour value $z_0 = 17/50$ in Section 11.
$\mathfrak{r}$	Coprimality modulus in (9.2) (written R in [1]); distinct from a counting exponent R .
$\mathcal{R}, \mathcal{M}$	Reciprocity phase; Mellin transform of the Fourier-transformed weight.
H_η, J_η, f_η	Nonvanishing correction; normalized average; principal-residue Mellin integral.
$\Gamma_1(3), \Gamma_2; \Gamma(s)$	Matrix groups; Euler gamma function.

Other indices and length parameters are local to the statement in which they are introduced. A *zero class* is a set of elements satisfying the same pair of zero-location inequalities (5.1); a *row* means one outer element index in a character-polynomial sum.

Characters and their presentations. For an integral modulus $\mathfrak{f}$, the ray class group is $I(\mathfrak{f})/P_1(\mathfrak{f})$: $I(\mathfrak{f})$ consists of fractional ideals coprime to $\mathfrak{f}$, and $P_1(\mathfrak{f})$ of principal ideals generated by elements congruent to 1 modulo $\mathfrak{f}$. A finite-order Hecke character is a homomorphism from this finite group to roots of unity. Its conductor is the smallest modulus through which it factors; it is primitive when the displayed modulus is its conductor. A presentation includes a possibly larger defining modulus and the zero extension at its nonunits, as in [1, Section 4]. Initially,

$$L_F(s, \eta) = \sum_{\mathfrak{a} \neq 0} \eta(\mathfrak{a})(N\mathfrak{a})^{-s} = \prod_p (1 - \eta(p)q_p^{-s})^{-1}, \quad \text{Re } s > 1;$$

outside this region the notation refers to its meromorphic continuation. A norm twist multiplies a character by $\mathfrak{a} \mapsto (N\mathfrak{a})^{it}$, $t \in \mathbb{R}$.

For a prime $p \notin S$, the sextic residue character is

$$\chi_p(a) = (a/p)_6, \quad \chi_p(a) \equiv a^{(q_p-1)/6} \pmod{p} \quad (p \nmid a), \quad \chi_p(a) = 0 \quad (p \mid a).$$

For primary $c = \prod p^{v_p(c)}$ put $\chi_c(a) = \prod_{p|c} \chi_p(a)^{v_p(c)}$. Throughout, all integer powers, including negative ones, are zero on nonunits, including powers divisible by six:

$$(2.1) \quad \chi_c(a)^6 = 1_{(a,c)=1}.$$

Passing to a primitive inducing character therefore requires retaining the removed coprimality conditions. Coefficients and such conditions are independent of the variable being averaged, except for the explicitly displayed residue symbols.

The additive normalization is

$$e(z) = \exp(2\pi i \operatorname{Tr}_{F/\mathbb{Q}}(z/\lambda)), \quad d\mu(z) = \frac{2}{\sqrt{3}} d\xi_1 d\xi_2.$$

Here $z = \xi_1 + i\xi_2$ with $\xi_1, \xi_2 \in \mathbb{R}$; for $z \in \mathbb{C}$ the trace in $e(z)$ means $z/\lambda + \overline{z}/\lambda$. The pairing $(z_1, z_2) \mapsto e(z_1 z_2)$ makes $\mathcal{O}$ self-dual. For squarefree c define

$$\gamma_j(c) = q_c^{-1/2} \sum_{v \bmod c} \chi_c(v)^j e(v/c).$$

Fixed-numerator symbols $n \mapsto \chi_n(u)$ are finite-order ray characters [1, Lemma 4.1]. Prime Gauss identities and sextic reciprocity provide finite-ray phases $\mathcal{R}, G$. Precisely, $\mathcal{R}$ is the bicharacter on residue square classes modulo 4 defined in [1, Lemma 4.4]; on coprime primary pairs it satisfies

$$\chi_b(a) = \mathcal{R}(a, b) \chi_a(b).$$

The same lemma defines the modulus-one finite-ray function G on all primary indices; on squarefree c , $G(c) = \overline{\chi_c(4)} \gamma_3(c)$. Its extension to nonsquarefree indices is that finite-ray function, rather than a Gauss sum, and satisfies

$$G(AB) = G(A)G(B)\mathcal{R}(A, B)$$

on the admissible primary elements [1, Lemmas 4.2–4.4]. These phases cancel cross-prime factors in the later Euler product.

3. A CRITERION FOR ZERO-FREE CONTINUATION

For a finite excluded set,

$$L_F^S(s, \eta) = L_F(s, \eta) \prod_{p \in S} (1 - \eta(p) q_p^{-s}).$$

The deleted factors are nonzero for $\operatorname{Re} s > 0$.

Proposition 3.1 (Continuation criterion, [1, Proposition 2.1]). *Fix $1/2 < \sigma_0 < 1$ and $C(s) = s + c$, $c \in \mathbb{R}$. Suppose that, whenever $\beta_* > \sigma_0$, the following construction is available. For every primitive η , suppose H_η is holomorphic on $\operatorname{Re} s > \sigma_0$ and*

$$(3.1) \quad \sup_{\operatorname{Re} s > \sigma_0} |H_\eta(s) - 1| \leq 1/2.$$

Define

$$(3.2) \quad f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds.$$

Here (v) denotes a vertical line of real part v . Assume there is a normalized average J_η satisfying

$$(3.3) \quad |J_\eta(Z)| \ll_\eta Z^{C(\sigma_0) + \omega_0}, \quad 0 < \omega_0 < \beta_* - \sigma_0,$$

$$(3.4) \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*) - \sigma}, \quad \sigma > 0,$$

The positive numbers ω_0, σ are independent of η ; constants, finite excluded sets and lower thresholds may depend on η . Then $\beta_* \leq \sigma_0$.

Idea of the continuation argument. Assume $\beta_* > \sigma_0$ and put

$$\epsilon_c = \min\{\beta_* - \sigma_0 - \omega_0, \sigma, (\beta_* - \sigma_0)/2\} > 0.$$

The two estimates give $f_\eta(Z) \ll_\eta Z^{C(\beta_*) - \epsilon_c}$ for $Z \geq 1$. Moving (3.2) to any fixed line $\operatorname{Re} s = b_c > 2$ gives $f_\eta(Z) \ll_{\eta, b_c} Z^{b_c + c}$ for $0 < Z \leq 1$. Consequently

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s-c} \frac{dZ}{Z}$$

is holomorphic on $\operatorname{Re} s > \beta_* - \epsilon_c > \sigma_0$. Mellin inversion and analytic continuation identify

$$F_\eta(s) = e^{(s-5/6)^2} H_\eta(s) / L_F^S(s, \eta) \quad (\operatorname{Re} s > 1) \quad [1, \text{Proposition 2.1}].$$

Since $|H_\eta| \geq 1/2$, the function

$$e^{-(s-5/6)^2} F_\eta(s) / H_\eta(s)$$

is a holomorphic continuation of $1/L_F^S(s, \eta)$ to $\operatorname{Re} s > \beta_* - \epsilon_c$, for every primitive η . The supremum definition supplies some η_0 and a zero ρ there; attainment of β_* is unnecessary. Locally,

$$L_F^S(s, \eta_0) = (s - \rho)^{k_\rho} g(s), \quad k_\rho \geq 1, \quad g(\rho) \neq 0.$$

Its reciprocal has a pole, contradicting this holomorphic continuation. Neither the Gaussian nor H_{η_0} can cancel it. Principal poles, in contrast, give zeros of the reciprocal and cause no obstruction.

The two applications in Part I use

σ_0	$C(s)$	$C(\sigma_0)$
11/12	$s - 2/3$	1/4
7/8	$s - 11/16$	3/16

and different normalized averages.

4. CUBIC THETA SERIES AND THE SMOOTHED AVERAGE

4.1. The cubic theta function and its Fourier coefficients. The function used here is *Kubota's cubic metaplectic theta function* θ_{cub} on $\mathbb{H}^3 = \mathbb{C} \times \mathbb{R}_{>0}$. Kubota developed the underlying metaplectic construction [2]; Patterson determined the explicit cubic theta coefficients [3, Theorem 8.1]. We use the normalization and cusp formulas recorded by Dunn–Radziwiłł [4, Section 5.1 and Appendix A], as implemented in [1, Proposition 5.1]. Here *cubic* refers to the cubic residue symbol in its automorphy multiplier. Set

$$\Gamma_1(3) = \{g \in \operatorname{SL}_2(\mathcal{O}) : g \equiv I \pmod{3}\}, \quad \Gamma_2 = \langle \operatorname{SL}_2(\mathbb{Z}), \Gamma_1(3) \rangle.$$

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \operatorname{SL}_2(\mathbb{C})$, its action on $w = (z, v) \in \mathbb{H}^3$ is

$$g(z, v) = \left(\frac{(az + b)\overline{(cz + d)} + a\bar{c}v^2}{|cz + d|^2 + |c|^2v^2}, \frac{v}{|cz + d|^2 + |c|^2v^2} \right).$$

The transformation law is

$$\theta_{\text{cub}}(gw) = \kappa_{\text{cub}}(g)\theta_{\text{cub}}(w), \quad \kappa_{\text{cub}}(g) = \begin{cases} (c/a)_3, & g \in \Gamma_1(3), \ c \neq 0, \\ 1, & g \in \Gamma_1(3), \ c = 0, \text{ or } g \in \operatorname{SL}_2(\mathbb{Z}). \end{cases}$$

The multiplier extends to Γ_2 as a homomorphism [3, Section 2]; the action and multiplier above are [4, equations (5.1)–(5.4)]. The symbol $(c/a)_3$ is the cubic residue symbol, multiplicative in its primary denominator; at good primes $(x/p)_3 = \chi_p(x)^2$. The notation κ_{cub} distinguishes this multiplier from the real zero-bound parameter κ .

The Fourier expansions needed in the proof are those at the three cusps represented by

$$H_j = \begin{pmatrix} 1 & 0 \\ t_j & 1 \end{pmatrix}, \quad (t_0, t_1, t_2) = (0, \omega^2, \omega), \quad j = 0, 1, 2.$$

Each H_j sends ∞ to a boundary cusp in $F \cup \{\infty\}$; expanding $\theta_{\text{cub}} \circ H_j$ gives its Fourier expansion there. With $\check{e}(x) = \exp(2\pi i(x + \bar{x}))$, these expansions are

$$\overline{\theta_{\text{cub}}(H_j(z, v))} = C_j(v) + \sum_{\substack{\mu \in \lambda^{-4}\mathcal{O} \\ \mu \neq 0}} d_j(\mu) v K_{1/3}(4\pi|\mu|v) \check{e}(\mu z).$$

Here

$$C_0(v) = \frac{3^{5/2}}{2} v^{2/3}, \quad C_1(v) = C_2(v) = 0.$$

The symbol d_j denotes the fixed arithmetic Fourier-coefficient function, and $K_{1/3}$ the modified Bessel function of the second kind. The precise correspondence with the notation of [4] is given in Appendix A. In particular, for primary squarefree c and primary n ,

$$d_0(\lambda^{-3}cn^3) = 3^{5/2}|n|\tilde{g}_3(c), \quad \tilde{g}_3(c) = q_c^{-1/2} \sum_{x \bmod c} (x/c)_3 \check{e}(x/c).$$

When $(c, S) = 1$ this becomes

$$\tilde{g}_3(c) = \chi_c(\lambda)^{-2} \gamma_2(c).$$

Thus the squarefree Gauss coefficients and the cube multiples in the following Dirichlet series come directly from these Fourier coefficients. The automorphy law supplies the reflection formula; its explicit coefficients supply the mean-square estimates. These are the theta inputs used below; Appendix A checks the constants directly from [4, equations (5.6), (5.7), (5.15), (5.16)].

4.2. Completed theta coefficients and mean squares. For a ray character Ψ with the conventions of Section 2, and a weight $V \in C^\infty(0, \infty)$ satisfying, for every $C > 0$ and $j \geq 0$,

$$(x\partial_x)^j V(x) = O_{C,j}(x^C) \quad (x \rightarrow 0), \quad (x\partial_x)^j V(x) = O_{C,j}(x^{-C}) \quad (x \rightarrow \infty),$$

put

$$\begin{aligned} D(s, \Psi) &= \sum_{c \text{ sf}} \gamma_2(c) \overline{\varpi(c)} \Psi(c) q_c^{-s}, \\ L_0(s, \Psi) &= \sum_n \overline{\varpi(n)}^3 \Psi(n)^3 q_n^{-3s+1/2}, \\ T(s, \Psi) &= D(s, \Psi) L_0(s, \Psi). \end{aligned}$$

The series $T(s, \Psi)$ is the character-twisted Dirichlet series built from the theta coefficients after the fixed rescaling and phase conversion in [1, Proposition 5.1]; it converges absolutely for $\operatorname{Re} s > 1$. Its smoothed expression is

$$(4.1) \quad \mathcal{C}_V(X; \Psi) = \sum_{c \text{ sf}, n} \frac{\gamma_2(c) \overline{\varpi(cn^3)} \Psi(cn^3)}{\sqrt{q_c} q_n} V(q_c q_n^3 / X).$$

No condition $(c, n) = 1$ is imposed. The cube factor is included because the natural theta support consists of the completed indices cn^3 . Here *completed* means that all cube multiples of the squarefree coefficient are included. For a primary A ,

$$A = cn^3 \iff v_p(A) \equiv 0 \text{ or } 1 \pmod{3} \quad \text{for every } p,$$

with c squarefree and c, n uniquely determined primary generators. It is an arithmetic completion of the indices; the gamma factors enter separately through reflection.

Proposition 4.1 (Cubic theta reflection, [1, Proposition 5.1]). *Fix a finite family of ray characters with conductors supported on S , and a common periodicity modulus L_{per} supported on S , divisible by 18 and their defining moduli, as specified in [1, Proposition 5.1, hypotheses]. Write $\Psi = \Psi_0 \prod_{p \in \mathcal{P}} \chi_p^{j_p}$ on primary elements, with $\mathcal{P}$ a finite set of distinct primes outside S and $j_p \in \{0, \dots, 5\}$; zero extensions are retained. The product $T(s, \Psi)$ continues to an entire function with polynomial growth on each fixed vertical strip, with constants depending on the fixed arithmetic data and the strip. For $X > 0$ and the test V specified above, the precise reflection identity is*

$$(4.2) \quad \mathcal{C}_V(X; \Psi) = \sum_{\mathfrak{t}} \zeta_{\mathfrak{t}} \sum_{0 \neq \mu \in \lambda^{-4}\mathcal{O}} \frac{d_{\mathfrak{t}}(\mu) \varpi(\mu) \vartheta_{\mathfrak{t}}(\lambda^4 \mu)}{\sqrt{q_{\mu}}} \prod_{p \text{ active}} B_p(\lambda^4 \mu) V^{\sharp}(q_{\mu} X / q_{c_{\mathfrak{t}}}^2).$$

Here $\mathfrak{t}$ runs over the terms indexed by $h_0 \bmod L_{\text{per}}$ and an active subset of $\mathcal{P}$ [1, Proposition 5.1, item (1)]. Every prime with $j_p \neq 0$ is active. The denominator is $c_{\mathfrak{t}} = c_F \prod_{p \text{ active}} p$, where c_F is the reduced denominator of $\lambda^2 h_0 / L_{\text{per}}$, with the normalizing unit fixed as in that item. The function $d_{\mathfrak{t}}$ is one of d_0, d_1, d_2 ; $\vartheta_{\mathfrak{t}}$ is an additive character modulo a fixed modulus depending on L_{per} ; the scalar $\zeta_{\mathfrak{t}}$ is independent of μ and has $|\zeta_{\mathfrak{t}}| \leq 1/81$. Their exact phases are specified in [1, Proposition 5.1, items (2)–(3)]. The local factors, all zero on nonunits when a character is evaluated, are

$$B_p(x) = \begin{cases} \chi_p(x)^{-j_p-2}, & j_p \neq 0, 4, \\ q_p^{-1/2}(-1 + q_p 1_{p|x}), & j_p = 4, \\ q_p^{-1/2} \chi_p(x)^{-2}, & j_p = 0 \text{ and } p \text{ active.} \end{cases}$$

Its radial transform is

$$V^\sharp(x) = \frac{1}{2\pi i} \int_{(v)} \widehat{V}(-t) \prod_{\pm} \frac{\Gamma(1+t \pm 1/6)}{\Gamma(1-t \pm 1/6)} \left(\frac{(2\pi)^4 x}{27} \right)^{-t} dt, \quad v \geq 0,$$

where Γ is the Euler gamma function and $\widehat{V}(t) = \int_0^\infty V(x) x^t dx/x$. The dual coefficients have support $u\lambda^k c n^3$, where $u \in \mathcal{O}^\times$ is a unit, $k \in \mathbb{Z}$, $k \geq -4$, c is squarefree, and c, n are primary. At such an index each of the three fixed cusp coefficient functions d_j of Section 4.1 has absolute value $\leq 27 3^{k/6} |n|$ [1, Proposition 5.1]. The formula retains the local character zeros, including the coprimality conditions of imprimitive presentations.

The quadratic large sieve over number fields is [5, Definition 1 and Theorem 1.1].

Lemma 4.2 (Completed theta mean square, [1, Lemma 5.8]). *Fix the arithmetic data, bounded ranges for M, N and dyadic support comparisons. For every $\varepsilon > 0$, there exist finite seminorm orders A, J such that, for all sufficiently large Z and every nonempty powerful-part dyadic range O , one has*

$$(4.3) \quad \sum_{\substack{0 < q_m \ll Z^M \\ q_{m_{\text{pow}}} \asymp Z^O}} |\mathcal{C}_V(Z^N; \nu \chi_\bullet(m))|^2 \ll Z^{O/2 + \max(M-O, 2M-O-N) + \varepsilon} \mathfrak{M}(V)^2.$$

Here $M, O \geq 0$ and $N \in \mathbb{R}$ are norm-range exponents, ν is a fixed finite ray character, and $\chi_\bullet(m)$ means $n \mapsto \chi_n(m)$. The ideal $m_{\text{pow}} = \prod_{v_p(m) \geq 2} p^{v_p(m)}$ is the powerful part. The seminorm is $\mathfrak{M} = \mathfrak{M}_{A,J}$ for sufficiently large fixed A, J , where [1, Proposition 5.1] defines

$$\mathfrak{M}_{A,J}(V) = \sup_{-A \leq \sigma \leq 1/4} \int_{\mathbb{R}} (1+|t|)^J |\widehat{V}(\sigma + it)| dt.$$

Fixed ray decompositions and kernel decay justify the specialization to the Gaussian test [1, Lemmas 5.2–5.7].

For the later divisibility restrictions, the corresponding mean-square estimate is [1, Lemma 14.3]. Its arithmetic input [1, Lemma 14.2] combines quadratic and cubic sieves.

Lemma 4.3 (Quadratic–cubic sieve, [1, Lemma 14.2]). *Let K, N, B, L be independent positive real norm scales:*

$$K, N, B, L \geq 1, \quad q_k \ll K, \quad q_n \asymp N, \quad q_b \asymp B, \quad q_P \asymp L.$$

Here k, n, P are squarefree primary ideals and b is any primary ideal. Thus $KNBL$ is the product of the four norm scales, not a further variable. The indices k, P avoid S ; n, b

may meet S except at λ , as specified in [1, Lemma 14.2]. For every $\varepsilon > 0$,

$$(4.4) \quad \sum_{k \text{ sf}, q_k \ll K} \left| \sum_{P \text{ sf}, q_P \asymp L} \frac{a(P)}{\sqrt{q_P}} 1_{(P,k)=1} \sum_{\substack{n \text{ sf}, q_n \asymp N \\ q_b \asymp B}} \beta(n, b) \chi_k(nb)^3 \chi_P(n)^{-2} 1_{(P,b)=1} \right|^2 \\ \ll (KNBL)^\varepsilon (K + NB) B \{N + L + (NL)^{2/3}\}.$$

The coefficient functions satisfy $|a(P)|, |\beta(n, b)| \leq 1$; $a(P)$ is independent of k, n, b , and $\beta(n, b)$ of k, P .

The ideal P need not be prime. Inclusion-exclusion removes the moving coprimality condition before applying the sieves. The full reflected estimate follows by dyadic decomposition, the local theta coefficients, and decay of $V^\sharp$.

4.3. Definition and separation. Choose a fixed finite ray group $\mathcal{T}$ for the supplementary phases [1, Lemma 4.4]. Here $\widehat{\mathcal{T}}$ is its character group, and $1_{\mathcal{T}}$ its identity ray class. Membership $p \in 1_{\mathcal{T}}$ means that the primary generator of p has trivial image in $\mathcal{T}$. Write $\Theta = \langle \eta, \widehat{\mathcal{T}} \rangle$ for the finite group of Hecke characters generated by these characters; all its conductor primes are in S [1, Section 8]. Let b_* generate the squarefree product of primes in S and choose a primitive character $\xi \bmod b_*$ of order dividing six, nonprincipal at each local prime. For a residue character ψ modulo a primary modulus c and $h \in \mathcal{O}$, set

$$g_\psi(c, h) = \sum_{d \bmod c} \psi(d) e(hd/c), \quad \tau_* = q_{b_*}^{-1/2} g_\xi(b_*, 1), \quad |\tau_*| = 1, \quad \Xi(A) = \xi(A) \chi_A(b_*).$$

For the primary averaging variable v let

$$\Psi_{m,v,\eta}(A) = \eta(A) \Xi(A)^{-1} \mathcal{R}(A, v) \chi_A(m)$$

on primary elements coprime to S , extended by zero elsewhere, and define

$$T_{m,v,\eta}(t) = \sum_{c \text{ sf}, n} \gamma_2(c) \overline{\varpi(cn^3)} \Psi_{m,v,\eta}(cn^3) \overline{G(cn^3)} q_c^{-1/2-t} q_n^{-1-3t}.$$

For fixed nonnegative nonzero annular W_0, W_1 and $\Phi(t) = e^{t^2}$, the source's average is

$$(4.5) \quad I_\eta(X, Y, Z) = \frac{1}{Y} \sum_v \frac{W_1(q_v/Y) \chi_v(b_*)}{\tau_* \xi(v) \sqrt{q_{b_*} q_v X}} \sum_m \xi(m) q_v^{-1/2} g_{\chi_v}(v, -m) \\ \times W_0(q_m/(q_{b_*} q_v X)) \frac{1}{2\pi i} \int_{(4)} Z^t \Phi(t) T_{m,v,\eta}(t) dt.$$

The m, v sums are finite, and the completed series converges absolutely on the initial line.

Expand $\overline{G(A)} = \sum_{\theta \in \widehat{\mathcal{T}}} a_\theta \theta(A)$, where $\sum |a_\theta| \leq |\mathcal{T}|^{1/2}$. For a class $\sigma \in \mathcal{T}$ and representative v_σ , put $\nu_\sigma(A) = \eta(A) \Xi(A)^{-1} \mathcal{R}(A, v_\sigma)$ and

$$B_{m,\sigma}(Z) = \sum_\theta a_\theta \frac{1}{2\pi i} \int_{(4)} Z^t \Phi(t) T(t + 1/2, \nu_\sigma \theta \chi_\bullet(m)) dt, \\ A_{m,\sigma,t_0}(Y) = \frac{1}{Y} \sum_{v \in \sigma} \frac{W_1(q_v/Y) \chi_v(b_*)}{\tau_* \xi(v)} (q_v/Y)^{-1/2+it_0} q_v^{-1/2} g_{\chi_v}(v, -m).$$

Writing $Q = q_{b_*} XY$, Mellin inversion separates the average:

$$(4.6) \quad I_\eta = \frac{Q^{-1/2}}{2\pi} \sum_{\sigma \in \mathcal{T}} \int_{\mathbb{R}} \widehat{W}_0(it_0) \sum_{m \neq 0} \Omega(q_m/Q) \xi(m) (q_m/Q)^{-it_0} A_{m,\sigma,t_0}(Y) B_{m,\sigma}(Z) dt_0.$$

Here $t_0 \in \mathbb{R}$ is the Fourier integration variable, $\widehat{W}_0(it_0) = \int_0^\infty W_0(r)r^{it_0} dr/r$, and Ω is a fixed smooth cutoff equal to one on the support of W_0 [1, Section 6]. Thus Cauchy–Schwarz reduces a direct estimate to two mean squares.

4.4. Poisson summation and the scalar Euler factors. Poisson summation in m removes zero frequency and frequencies meeting S , by primitivity of ξ . Every remaining element is uniquely ua^6 , where u includes its unit and has prime valuations at most five, while a is primary. Let $\widehat{W}_0(|y|^2)$ be the planar Fourier transform of $W_0(|y|^2)$ with the pairing $e(xy)$ and measure $d\mu$ defined in Section 2; this is the transform in [1, Section 7]. Set

$$\mathcal{M}(z) = \int_0^\infty \widehat{W}_0(r)r^{z-1}dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r)r^{w-1}dr,$$

$$\mathcal{W} = X^{1/2-z}Z^{s+z-1}Y^{w-1}\Phi(s+z-1)\mathcal{M}(z)\widehat{W}_1(w).$$

The complete local calculation [1, Lemma 7.1] gives

$$(4.7) \quad I_\eta = \frac{1}{(2\pi i)^3} \int_{(3)} \int_{(3)} \int_{(2)} \mathcal{W} \sum_u^{(6)} q_u^{-z} \overline{\xi(u)} \frac{\zeta_F^S(6z) L_F^S(w, \chi_\bullet(u))}{L_F^S(s, \eta \chi_\bullet(u))} \mathcal{H}_{\eta,u}(s, w, z) dz dw ds.$$

Here $\chi_\bullet(u) : n \mapsto \chi_n(u)$ and $\sum^{(6)}$ means sixth-power-free nonzero elements coprime to S .

Let $\zeta_F^S(s) = L_F^S(s, 1)$ be the Dedekind zeta function with the primes of S deleted. The local coefficient $P_p = P_p(s, w, z; \eta, u)$ is the complete valuation series computed in [1, Lemma 7.1], immediately before and in its local identity: $v_p(c) \in \{0, 1\}$ and $v_p(n), v_p(v), v_p(a) \geq 0$ are summed, with $v_p(u) \in \{0, \dots, 5\}$ fixed. This summary's s is the variable called x in that lemma. For its scalar Euler factors write

$$V_p = q_p^{-6z}, \quad W_p = \chi_p(u)q_p^{-w}, \quad D_p = \eta(p)\overline{\chi_p(u)}q_p^{-s}, \quad H_p = P_p \frac{(1 - V_p)(1 - W_p)}{1 - D_p}.$$

Then $\mathcal{H}_{\eta,u} = \prod_{p \notin S} H_p$. Gauss identities, reciprocity and the inserted phases factor the coefficient into these local terms. The six cases $v_p(u) = 0, \dots, 5$ are summed before extracting the scalar quotient. The correction is holomorphic in the two regions

$$(4.8) \quad \mathcal{D}_1 : \quad s_r \geq 51/100, \quad z_r \geq 17/50, \quad w_r \geq -1/100, \quad s_r + w_r \geq 1 + \varepsilon_0,$$

$$(4.9) \quad \mathcal{D}_2 : \quad s_r \geq 7/8, \quad z_r \geq 33/200, \quad w_r \geq 19/20,$$

where $s_r = \operatorname{Re} s$, $w_r = \operatorname{Re} w$, $z_r = \operatorname{Re} z$, and $\varepsilon_0 > 0$ is a fixed margin, with $\mathcal{H}_{\eta,u} \ll_\varepsilon q_u^\varepsilon$ uniformly in heights.

The only principal numerator is $u = 1$: a good prime in u forces nontrivial ramification, and a nontrivial unit gives a nontrivial Kummer character. At $u = 1$ the poles are $w = 1, z = 1/6$. Define

$$(4.10) \quad H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6), \quad c_S = \widehat{W}_1(1)\mathcal{M}(1/6) \frac{(\operatorname{Res}_{v=1} \zeta_F^S(v))^2}{6} > 0.$$

The $1/6$ is the residue Jacobian. A positive Gaussian-integral representation proves $\mathcal{M}(1/6) > 0$. Since $(s, 1, 1/6) \in \mathcal{D}_2$ for $\operatorname{Re} s > 7/8$, (4.9) and (4.10) give holomorphy of H_η there. The uniform local estimate $H_\eta(s) = 1 + O(P_0^{-4/5})$ [1, Lemma 7.1] allows P_0 to be chosen sufficiently large that

$$(4.11) \quad \sup_{\operatorname{Re} s > 7/8} |H_\eta(s) - 1| \leq 1/2.$$

The cutoff is independent of the target phases; see also [1, Section 11, principal correction].

5. ZERO DETECTION BY TWO CHARACTER POLYNOMIALS

For the zero-detection method, see [8, Appendix C, Zero detecting setup]. The simultaneous Hecke-character version used here is [1, Proposition 8.3].

For $q_u \asymp U = Z^d$ consider

$$\mathcal{X}_u = \{\nu(n)\chi_n(u)^{\pm 1} : \nu \in \Theta\}.$$

For a presentation ψ , let ψ^* denote its primitive inducing character, and $L(s, \psi)$ its Hecke series with the original zero extension; thus $L(s, \psi^*) = L_F(s, \psi^*)$. Primitive conductors have norm $\ll_{\mathcal{A}} U$; original presentations retain Euler deletions supported on S and u . Choose $0 < \epsilon_b < 10^{-3}$ and $T_1 = Z^\tau \leq U^{1/100}$. Set $I_{\epsilon_b} = \lceil 1/\epsilon_b \rceil + 2$. With $a_0 = 51/100$ and $j \in \{1, \dots, I_{\epsilon_b}\}$, let

$$M_j(u) = \max(\{a_0\} \cup \{\operatorname{Re} \rho : \psi \in \mathcal{X}_u, L(\rho, \psi^*) = 0, \operatorname{Re} \rho \geq a_0, |\operatorname{Im} \rho| \leq 3jT_1\}).$$

A pigeonhole argument gives $1 \leq i < I_{\epsilon_b}$ and $a \in (a_0 + \epsilon_b \mathbb{Z}_{\geq 0}) \cap [a_0, 1]$ such that

$$(5.1) \quad a \leq M_i(u) < a + \epsilon_b, \quad M_{i+1}(u) < a + 2\epsilon_b \quad [1, \text{Lemma 8.1}].$$

Hence the associated functions have a zero-free rectangle to the right of $a + 2\epsilon_b$ at the needed height. Concretely,

$$\mathcal{U}_{a,i} = \{s \in \mathbb{C} : a + 2\epsilon_b \leq \operatorname{Re} s \leq 2, |\operatorname{Im} s| \leq 3(i+1)T_1\}$$

contains no zero of $L_F(s, \psi^*)$ for $\psi \in \mathcal{X}_u$, by (5.1) and the definition of M_{i+1} . If $a > a_0$, a selected zero $\rho = \sigma + i\gamma$ satisfies $a \leq \sigma < a + \epsilon_b$, $|\gamma| \leq 3iT_1$. Since these are actual family zeros, $a \leq \beta_*$.

Let $\mathcal{B}$ denote a subset of sixth-power-free elements $q_u \asymp U$ in one fixed (i, a) class; later it may also fix prime-sum size ranges. All class counts are finite for fixed parameters [1, Lemma 8.1]. For one presentation $\psi \in \mathcal{X}_u$, real length exponents $r, m \geq 0$, and annular weights W_M, W_S , define

$$M_r = U^{-r/2} \sum_n \mu(n)\psi(n)W_M(q_n/U^r), \quad S_m = U^{-m/2} \sum_n \psi(n)W_S(q_n/U^m).$$

The profiles may contain pure norm twists. Polynomial bounds on fixed vertical strips, logarithmic bounds in zero-free disks and deleted Euler estimates [1, Lemmas 4.8–4.10] yield

$$(5.2) \quad |M_r|^2 \ll U^{\delta r + \varepsilon}, \quad |S_m|^2 \ll U^{\delta \min(m, 1-m) + \varepsilon}, \quad \delta = 2a - 1,$$

when $(1+T_1)^A \leq U^{\varepsilon/10}$ [1, Lemma 8.2]. The second estimate takes the better of the direct and reflected Hecke functional-equation bounds.

Proposition 5.1 (Simultaneous large values, [1, Proposition 8.3]). *Fix $\varepsilon > 0$, a class (i, a) from (5.1) with $a > a_0$, and a sixth-power-free u of norm $\asymp U$ in that class. Assume $T_1 \leq U^{1/100}$ and the height bound $(1+T_1)^A \leq U^{\varepsilon/10}$ from [1, Lemma 8.2], after choosing its preliminary losses sufficiently small. For every $t \in [1, 3/2]$ and all $U \geq U_0(\mathcal{A}, \varepsilon)$, there exist a common presentation $\psi \in \mathcal{X}_u$, a selected zero ρ of its primitive inducing function, and two polynomials with the same presentation and twist height such that*

$$(5.3) \quad r \leq t + O(1/\log U), \quad r + m \geq t - O(1/\log U),$$

$$(5.4) \quad t - \frac{1}{2} - O(\varepsilon) \leq r, \quad 0 \leq m \leq \frac{1}{2} + O(\varepsilon),$$

$$(5.5) \quad |M_r|^2 \gg U^{\delta r - \varepsilon}, \quad |S_m|^2 \gg U^{\delta m - \varepsilon} \quad [1, \text{Proposition 8.3}].$$

Idea of zero detection. To obtain them, put

$$C_\psi(s) = \sum_l \mu(l) \psi(l) V_\leq(q_l/U^t) q_l^{-s}, \quad V_\leq = 1 \text{ on } [0, 1], \quad V_\leq = 0 \text{ on } [2, \infty),$$

where $V_\leq$ is a fixed smooth cutoff. Here l ranges over integral ideals and t is the real truncation exponent. Use this truncated reciprocal in

$$\frac{1}{2\pi i} \int_{(2)} U^{20z} \Gamma(z) L(\rho + z, \psi) C_\psi(\rho + z) dz.$$

Shifting to $-1/4$ gives a small integral; the residue at zero vanishes because $L(\rho, \psi) = 0$. Expansion of LC_ψ cancels the ideal coefficients up to length U^t except the unit term. A remaining dyadic product therefore has $|M_r S_m|^2 \gg U^{\delta(r+m)-\varepsilon}$. Comparison with (5.2) forces $m \leq 1/2 + O(\varepsilon)$ and then both individual lower bounds. At $a = a_0$ there need be no selected zero; that class is counted directly.

6. THE PRELIMINARY 11/12 BOUND

Take $X = Y = Z^{1/2}$ and $Q = q_{b_*} Z$. The number-field additive large sieve originates in [6]; its exact planar form and application here are [1, Lemmas 6.1–6.2]. It gives

$$\sum_{q_m \ll Q} |A_{m, \sigma, t_0}(Y)|^2 \ll (Q + Y^2)/Y \ll Q/Y \quad [1, \text{Lemmas 6.1–6.2}].$$

Equation (4.3), at $M = N = 1$, gives $\sum |B_{m, \sigma}(Z)|^2 \ll Z^{1+\varepsilon}$. Thus (4.6) implies

$$(6.1) \quad |I_\eta(Z^{1/2}, Z^{1/2}, Z)| \ll Q^{-1/2} (Q/Y)^{1/2} Z^{1/2+\varepsilon} = Z^{1/4+\varepsilon} \quad [1, \text{Proposition 6.3}].$$

The antecedents are the cubic sieve [7, Theorem 2], the quadratic sieve [5, Theorem 1.1], and the higher-order norm recursion [9, Theorem 1.3 and Section 3].

Lemma 6.1 (Sextic sieve, [1, Lemma 9.1]). *Let $K, N_s \geq 1$ bound the norms of squarefree primary elements k, n coprime to S , and let c_n be finitely supported and independent of k . For every $\varepsilon > 0$,*

$$\sum_{k \text{ sf}, q_k \leq K} \left| \sum_{n \text{ sf}, q_n \leq N_s} c_n \chi_n(k)^{\pm 1} \right|^2 \ll (KN_s)^\varepsilon \{K + N_s + (KN_s)^{2/3}\} \sum_n |c_n|^2.$$

Proposition 6.2 (Preliminary character count, [1, Proposition 9.2]). *Let $\mathcal{B}$ be a zero class of sixth-power-free elements $q_u \asymp U$ as in Section 5, with the detector height assumptions and $\delta = 2a - 1$. For every $\varepsilon > 0$ and sufficiently large U ,*

$$(6.2) \quad \#\mathcal{B} \ll U^{R_1(\delta)+\varepsilon} (1 + T_1)^A, \quad R_1(\delta) = \min\{1, \max(1 - \delta/2, 4/3 - \delta)\} \quad [1, \text{Proposition 9.2}].$$

Proposition 6.3 (Preliminary zero-free bound, [1, Theorem 3.1]). *The family supremum in (1.2) satisfies*

$$(6.3) \quad \beta_* \leq 11/12.$$

The preliminary exponent comparison. Assume $\beta_* > 11/12$ and put $\Delta_1 = \beta_* - 11/12 > 0$. The principal residue in (4.7) has exponent $X^{1/3} Z^{s-5/6} = Z^{s-2/3}$. Hence

$$C_1(s) = s - 2/3, \quad J_{1, \eta}(Z) = I_\eta(Z^{1/2}, Z^{1/2}, Z)/c_S,$$

with $f_{1, \eta}$ defined by (3.2). The preliminary character count, the contour estimates [1, Lemmas 10.2–10.6], and the height choice in [1, Proposition 11.2] give

$$|J_{1, \eta} - f_{1, \eta}| \ll_\eta Z^{C_1(\beta_*)-1/4800}.$$

Apply Proposition 3.1 with

$$\sigma_0 = 11/12, \quad c = -2/3, \quad \omega_0 = \Delta_1/2, \quad \sigma = 1/4800.$$

The correction hypotheses follow from (4.9), (4.10) and (4.11). The displayed comparison bound and (6.1), with its loss chosen below $\Delta_1/2$, verify the two estimates, since $C_1(11/12) = 1/4$. Thus the preliminary argument reaches the contradiction through the same continuation criterion used for the $7/8$ bound.

7. THE MODIFIED AVERAGE AND ITS DIRECT ESTIMATE

Assume $\beta_* > 7/8$ and use (1.3). The averaging scales are

$$(7.1) \quad \begin{aligned} b = 1/8, \quad h = 13/16, \quad \ell = 1/6, \quad l_x = 17/48, \quad l_y = 23/48, \\ X = Z^{l_x}, \quad Y = Z^{l_y}, \quad l_x + l_y = 5/6, \quad b = l_y - l_x = 1/8, \quad h = 1 - l_x + \ell. \end{aligned}$$

Choose a positive integer K and K disjoint prime intervals with smooth nonnegative weights W_i , norms $q_p \asymp P_i = Z^{\ell_i}$, and $\sum_i \ell_i = \ell$. The allowed set $\mathcal{P}_i$ also imposes $p \notin S$ and $p \in 1_{\mathcal{T}}$. The underlying intervals are disjoint before these restrictions.

For a squarefree ideal D outside S , let $I_{\eta;D}$ denote (4.5) with $1_{D|cn^3}$ inserted. Write $J^c = \{1, \dots, K\} \setminus J$ and $p_J = \prod_{i \in J} p_i$ (empty product 1). All tuple sums have $p_i \in \mathcal{P}_i$. Define

$$(7.2) \quad \begin{aligned} I_{\eta, \text{mod}}(Z) = \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \\ \times I_{\eta; p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \end{aligned}$$

The one-prime operation is $\overline{\eta(p)} I_{\eta; p}(X, Y, Zq_p) - q_p^{-3/2} I_{\eta}(X/q_p, Y/q_p, Z)$. Its purpose is to cancel an unwanted scalar Euler contribution, leaving a sextic-character prime sum. All prime intervals retain their original scales.

Proposition 7.1 (Modified direct bound, [1, Proposition 15.3]). *For the fixed arithmetic data, prime intervals and smooth weights specified above, with the scales (7.1), one has*

$$(7.3) \quad |I_{\eta, \text{mod}}(Z)| \ll_{\eta, \varepsilon} Z^{3/16 + \varepsilon}.$$

For a fixed J put

$$d_J = \sum_{i \in J} \ell_i, \quad X' = X/q_{p_J}, \quad Y' = Y/q_{p_J}, \quad M' = 5/6 - 2d_J, \quad Q = q_{b_*} X' Y'.$$

Let $B_{m, \sigma}^J$ be the completed factor in the separated expression, with the primes $i \notin J$ already summed and their divisibility restrictions imposed. The theta functional equation with the divisibility restrictions [1, Corollary 14.1] and its reflected mean square give the following input.

Lemma 7.2 (Modified theta mean square, [1, Lemma 15.1]). *For every $\varepsilon > 0$, fixed subset J and ray class σ , uniformly in the fixed rescaled prime tuple,*

$$(7.4) \quad \sum_{q_m \ll Q} |B_{m, \sigma}^J|^2 \ll Z^{M' + ((d_J - 1/6)/4)_+ + \varepsilon} = Z^{M' + \varepsilon} \quad [1, \text{Lemma 15.1}].$$

The extra term vanishes because $0 \leq d_J \leq \ell = 1/6$. The remaining prime sums are included in $B_{m, \sigma}^J$.

Proposition 7.3 (Additive mean square, [1, Proposition 15.2]). *Fix $S, \mathcal{T}, b_*, \xi, W_1$ and a ray class σ . Let $Q, Y' \geq 1$ range over fixed polynomial ranges in Z , and suppose $P_a = Y'^2/Q \geq 1$. For every $\varepsilon > 0$ and real t_0 , with the coefficient A defined in Section 4,*

$$(7.5) \quad \sum_{q_m \ll Q} |A_{m, \sigma, t_0}(Y')|^2 \ll (1 + |t_0|)^{J_{\text{Gr}}} \frac{Q}{Y'} \left(1 + P_a^{1/6} + \frac{P_a^2}{Y'}\right) Z^\varepsilon \quad [1, \text{Proposition 15.2}].$$

Here J_{Gr} is a fixed seminorm order, independent of the subset J , moving scales and t_0 .

Idea of the additive estimate. Expansion of the square and Poisson summation reduce (7.5) to complete finite correlations. The common-factor identities and local prime-power formulas are [1, Lemmas 13.2–13.4]; the final mean-square estimate is [1, Proposition 15.2]. Both error terms $P_a^{1/6}$ and P_a^2/Y' in (7.5) are retained. Their comparison at the chosen scales is the calculation needed below.

The direct exponent calculation. With the fixed scales (7.1),

$$P_a = q_{b_*}^{-1} Z^{1/8}, \quad l_y - d_J - 11b/6 \geq 1/12, \quad P_a^2/Y' \ll P_a^{1/6}.$$

Thus $P_a \geq 1$ once $Z^{1/8} \geq q_{b_*}$, which holds for all sufficiently large Z with the arithmetic data fixed. Equations (7.4)–(7.5) and Cauchy–Schwarz therefore bound each separated term by $(X')^{1/2} P_a^{1/12} Z^\varepsilon$. The rescaled prime tuples have total weight

$$Z^{d_J + \varepsilon} O(Z^{-3d_J/2}), \quad (X')^{1/2} \asymp X^{1/2} Z^{-d_J/2}.$$

Consequently the exponent in (7.3) is $l_x/2 + b/12 = 3/16$, since $d_J - 3d_J/2 - d_J/2 = -d_J \leq 0$. The asymmetry in (7.1) was chosen to obtain this exponent while keeping the additive error under control.

8. LOCAL CANCELLATION AND PRINCIPAL RESIDUES

8.1. Holomorphic correction and local errors. Let P_p^* be the part of P_p with positive valuation of the completed index. For $Q_p = q_p$, the selected-prime operation replaces P_p by

$$Q_p^{z-1} \{\overline{\eta(p)} Q_p^s P_p^* - Q_p^{-w} P_p\}.$$

After extracting the same scalar Euler factors as in (4.7), define

$$(8.1) \quad G_p = \{\overline{\eta(p)} Q_p^s P_p^* - Q_p^{-w} P_p\} \frac{(1 - V_p)(1 - W_p)}{1 - D_p}.$$

The local identity of [1, Lemma 7.1] shows that G_p continues holomorphically in $\mathcal{D}_1, \mathcal{D}_2$. It does not require division by P_p . The correction is

$$(8.2) \quad \mathfrak{H}_{\eta, u, Z} = \sum_{(p_i)} \prod_i [W_i(q_{p_i}/P_i) q_{p_i}^{z-1} G_{p_i}] \prod_{\substack{p \notin S \\ p \notin \{p_i\}}} H_p.$$

Replacing $\mathcal{H}$ by $\mathfrak{H}$ in (4.7) gives the exact modified average.

On the contours used for the nonprincipal contributions, $s_r = a + 16\epsilon_b$, $w_r = 1 - a - 6\epsilon_b$, $z_r = 17/50$, [1, Section 16] decomposes

$$\mathfrak{H} = \mathcal{H} \prod_i (\mathcal{Q}_i + \mathcal{D}_i), \quad \mathcal{Q}_i = - \sum_{p \in \mathcal{P}_i} W_i(q_p/P_i) q_p^{z-1} \overline{\chi_p(u)}.$$

Precisely, on these contours define

$$\mathcal{B}_i = \sum_{p \in \mathcal{P}_i} W_i(q_p/P_i) q_p^{z-1} \frac{G_p}{H_p}, \quad \mathcal{D}_i = \mathcal{B}_i - \mathcal{Q}_i.$$

Proposition 8.1 (Joint local error, [1, Proposition 16.1]). *Let $a_0 \leq a \leq 1$, $0 < \epsilon_b \leq 10^{-3}$, $q_u \asymp U$, with u sixth-power-free and $(u, S) = 1$. Put $\delta = 2a - 1$. Suppose the primitive character ψ_u^* inducing $\chi_\bullet(u)$ is nonprincipal. On the displayed contours restrict to $|\Im w| \leq T_1$, and assume $|L_F(1 - w, \overline{\psi_u^*})| \ll U^\epsilon$ uniformly on the points under consideration. For sufficiently large P_0 , the local H_p are nonzero on these contours. Moreover, for every subset $I \subseteq \{1, \dots, K\}$, the joint error estimate is*

$$(8.3) \quad \left| L_F^S(w, \chi_\bullet(u)) \prod_{i \in I} \mathcal{D}_i \right| \ll U^{\delta/2 + O(\epsilon_b) + \epsilon} (3 + T_1)^A \prod_{i \in I} P_i^{17/50 - 1/2 + O(\epsilon_b) + \epsilon} \quad [1, \text{Proposition 16.1}].$$

Let $\mathfrak{f}_u$ be the conductor ideal of ψ_u^* , and I_0 any set of distinct selected primes dividing u . With $j_p = v_p(u) \in \{1, \dots, 5\}$,

$$q_{\mathfrak{f}_u} \ll_S q_{\text{rad}(u)} \leq q_u \prod_{p \in I_0} q_p^{-(j_p - 1)}.$$

Applying this conductor deficit once to the primitive numerator's functional equation pays for all strict ramified local errors simultaneously. Absolute bounds on the other contours use [1, Lemma 16.2].

8.2. The principal coefficient and the normalized average. At $u = 1$ in $\mathcal{D}_2$, $G_p/H_p = -1 + O(q_p^{-7/8})$ uniformly in heights.

Lemma 8.2 (Prime asymptotic in a fixed ray class, [1, Lemma 13.1]). *Fix the ray quotient $\mathcal{T}$, the excluded set S , and a nonnegative nonzero smooth annular weight W_i . As $P_i \rightarrow \infty$, with $\mathcal{P}_i$ restricted to the identity class of $\mathcal{T}$ and primes outside S , one has*

$$(8.4) \quad S_i(Z) := \sum_{p \in \mathcal{P}_i} W_i(q_p/P_i) q_p^{-5/6} \sim \frac{P_i^{1/6}}{|\mathcal{T}| \log P_i} \int_0^\infty W_i(y) y^{-5/6} dy > 0.$$

Fixed-ray prime counting follows by applying Chebotarev to the fixed ray-class field; see [10, Theorem 1.1]. Smooth partial summation gives (8.4).

For large Z set

$$(8.5) \quad A_{\mathcal{T}}(Z) = (-1)^K Z^{-\ell/6} \prod_i S_i(Z), \quad J_{\eta}(Z) = \frac{I_{\eta, \text{mod}}(Z)}{c_S A_{\mathcal{T}}(Z)}.$$

Then $|A_{\mathcal{T}}| \asymp (\log Z)^{-K}$ and $|A_{\mathcal{T}}|^{-1} \ll_\epsilon Z^\epsilon$, so

$$(8.6) \quad |J_{\eta}(Z)| \ll_{\eta, \epsilon} Z^{3/16 + \epsilon}.$$

At $w = 1, z = 1/6$ the correction equals

$$H_{\eta}(s) Z^{\ell/6} A_{\mathcal{T}}(Z) \{1 + O(Z^{-\kappa_P})\}, \quad 0 < \kappa_P < (7/8) \min_i \ell_i.$$

The residue exponent is

$$(8.7) \quad C(s) = s + l_x/3 - 5/6 + \ell/6 = s - 11/16, \quad C(7/8) = 3/16,$$

and the resulting integral is

$$(8.8) \quad f_{\eta}(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{s-11/16} e^{(s-5/6)^2} \frac{H_{\eta}(s)}{L_F^S(s, \eta)} ds.$$

Crossing $w = 1$ on the way to $19/20$, then $z = 1/6$ on the way to $33/200$, gives principal remainder exponents

$$C(\beta_*) + (1 + h)\epsilon_b - m_w + \epsilon, \quad C(\beta_*) + \epsilon_b - m_z + \epsilon, \quad C(\beta_*) + \epsilon_b - \kappa_P + \epsilon,$$

where

$$m_w = 23/960, \quad m_z = 13/9600 \quad [1, \text{Lemma 10.5}].$$

The reciprocal stays on $s_r = \beta_* + \epsilon_b$ throughout.

9. THE TWO MOMENT ESTIMATES

All lengths in this section are logarithmic exponents in a common base Z . Products of auxiliary prime polynomials use disjoint prime intervals and one common character presentation. For example, length r means norms $\asymp Z^r$. A mean square is a sum $\sum_u |F_u|^2$; the fourth moment arises from $\sum_u |S_u S_u|^2 = \sum_u |S_u|^4$.

9.1. Möbius-weighted second moment. Let $\psi_u(n) = \nu(n)\chi_n(u)^{\pm 1}$, with fixed ray character ν , and

$$M_u(Z^r) = Z^{-r/2} \sum_n \mu(n) \psi_u(n) W(q_n/Z^r), \quad Q_u = \prod_i Z^{-z_i/2} \sum_{p \in \mathcal{P}_i} a_i(p) \psi_u(p) W_i(q_p/Z^{z_i}).$$

Lemma 9.1 (Möbius-weighted moment, [1, Lemma 17.1]). *Let $r, m_0, z_i \geq 0$ be norm exponents: the ideal variable n has norm $\asymp Z^r$, u ranges up to Z^{m_0} , and the i th prime set has norms $\asymp Z^{z_i}$. The coefficients $a_i(p)$ are bounded and independent of u . Fix bounded ranges for these lengths and the number of prime factors. Writing $z_p = \sum_i z_i$, assume for a fixed $c_m > 0$ that*

$$r + 2z_p \leq m_0 - c_m, \quad 2r + 8z_p \leq 3m_0 - c_m.$$

Then for every $\varepsilon > 0$ and all sufficiently large Z ,

$$(9.1) \quad \sum_{0 < q_u \ll Z^{m_0}} |M_u(Z^r) Q_u|^2 \ll Z^{m_0 + \varepsilon} \quad [1, \text{Lemma 17.1}].$$

The implied constant may depend on c_m , the bounded ranges and finitely many smooth seminorms, but is uniform in prime supports and bounded coefficients. There is no restriction on individual prime lengths in this lemma.

Structure of the inverse-moment induction. One initial Poisson transform reduces the assertion to [1, Lemma 17.2]:

$$(9.2) \quad Z^{-V} \sum_{f \text{ sf}, q_f \asymp Z^V} w(f) \sum_{0 < q_k \ll Z^M} \left| Z^{-N/2} \sum_{n \text{ sf}} \overline{\varpi(n)} \gamma_2(n) \nu(n) 1_{(n, \mathfrak{r})=1} \chi_n(k f^4) \mathfrak{d}(n) W(q_n/Z^N) \right|^2 \ll Z^{F_0 + \varepsilon}, \quad F_0 = N + V,$$

under

$$q_{\mathfrak{r}} \leq Z^{F_0 - M - z_{\text{cap}} - c_*}, \quad 3M + 6z_{\text{cap}} + c_* \leq 4F_0, \quad c_* > 0.$$

The parameters $M, N, V, z_{\text{cap}} \geq 0$ in this lemma are local norm exponents: f has norm $\asymp Z^V$, k norm $\ll Z^M$, and n norm $\asymp Z^N$. The squarefree ideal $\mathfrak{r}$ is the coprimality modulus, independent of k, f ; $c_* > 0$ is a fixed margin and z_{cap} a cap on total prime length. Here $w(f)$ depends on f alone and is nonnegative and bounded by a fixed power of the divisor function. Also,

$$\mathfrak{d}(n) = \sum_{(p_i)} \prod_i a_i(p_i) 1_{p_i | n}$$

has total prime length at most z_{cap} : its tuples run over disjoint fixed prime sets of norms $\asymp Z^{z_i}$, with $\sum_i z_i \leq z_{\text{cap}}$. The coefficients a_i are bounded, with data independent of k, f . No additional arbitrary coefficient depending on k is admitted.

The inequalities in (9.1) supply the initial positive margins. Two finite Poisson transforms then decrease M while preserving the common character, coprimality condition and $4F_0 - 3M - 6z_{\text{cap}}$ margin [1, Lemma 17.3]. The reflected mean square [1, Lemma 14.3] handles the terminal range, and [1, Corollary 17.4] bounds the seminorm and height orders. These preserved margins are the features needed for the later counts.

Lemma 9.2 (Sixth-power amplification, [1, Lemma 17.6]). *Fix a ray character ν , a common sign $\varepsilon_\chi \in \{-1, 1\}$, and a smooth annular weight W , with the zero extensions of Lemma 9.1. Put $\psi_u(n) = \nu(n)\chi_n(u)^{\varepsilon_\chi}$ and use M_u as defined above, without auxiliary prime factors. Let $r \geq 0$ range over a fixed bounded interval and $q_u \asymp U$, with u sixth-power-free. For every $\varepsilon > 0$ and sufficiently large U ,*

$$(9.3) \quad \sum_{q_u \asymp U}^{(6)} |M_u(U^r)|^2 \ll U^{a(r)+\varepsilon} (1+T_1)^A, \quad a(r) = \max\{1, (1+5r)/6\} \quad [1, \text{Lemma 17.6}].$$

The estimate permits a separate row profile $W(y)y^{-\sigma_u+it_u}$, with σ_u in a fixed compact interval and $|t_u| \leq T_1$. The factor $(1+T_1)^A$ is its smooth-seminorm cost; A depends only on the fixed ranges and arithmetic data.

Idea of sixth-power amplification. For a norm scale $D_{\text{amp}} \geq 1$ and primary a , inclusion-exclusion gives

$$M_u(D_{\text{amp}}) = \sum_{d|\text{rad } a} \mu(d)\psi_u(d)q_d^{-1/2}M_{ua^6}(D_{\text{amp}}/q_d).$$

Average over $q_a \leq P_{\text{amp}} = (H_{\text{amp}}/U)^{1/6}$ with $H_{\text{amp}} = \max(2U, D_{\text{amp}}^{1+c})$. The map $(u, a) \mapsto ua^6$ is injective on sixth-power-free u , and the enlarged mean square costs $H_{\text{amp}}/P_{\text{amp}} = U^{1/6}H_{\text{amp}}^{5/6}$. Here $c > 0$ is a small fixed enlargement parameter. Choosing c small proves (9.3), including $r = 1$ without a shrinking strict margin.

9.2. Fourth moment of unweighted character polynomials.

Lemma 9.3 (Unweighted product moment, [1, Lemma 18.1]). *For $\psi_k(n) = \nu_{\text{fix}}(n)\chi_n(k)$ let m_0 be the element range exponent, and q_0 bound the radical of the displayed variable residue-symbol factors, meaning that the product of their distinct supporting primes has norm $\leq Z^{q_0}$. The twist ν_{fix} is fixed during the k -sum. Put $M = m_0 + q_0$. Allow a common coprimality condition with modulus of norm $O(Z^B)$ for a fixed B , independent of k and common to every factor. Prime coefficients are fixed finite combinations of characters in Θ . For $\psi = \psi_k$ use the definitions*

$$S_\psi(n_j) = Z^{-n_j/2} \sum_a \psi(a)W_j(q_a/Z^{n_j}), \quad Q_k = \prod_i Z^{-z_i/2} \sum_{p \in \mathcal{P}_i} a_i(p)\psi_k(p)W_i(q_p/Z^{z_i}),$$

where $n_1, n_2 \geq 0$ are length exponents, not ideals, and $z_p = \sum_i z_i$. Every factor includes the common coprimality indicator just specified. All real length parameters range over prescribed bounded sets. Let $3/4 \leq \kappa \leq 1$; if $z_p > 0$ and $\kappa < 1$, assume $\beta_ \leq (1+\kappa)/2$. For every $\varepsilon > 0$ there exists $\ell_{\text{mesh}} > 0$, depending only on these ranges and ε , such that $z_i \leq \ell_{\text{mesh}}$ for every i implies*

$$(9.4) \quad \sum_{k \in \mathcal{R}_{z_p}} |S_{\psi_k}(n_1)S_{\psi_k}(n_2)Q_k|^2 \ll Z^{M+\varepsilon}.$$

Here $\mathcal{R}_{z_p}$ consists of the elements $0 < q_k \ll Z^{m_0}$ whose primitive inducing characters satisfy the exclusions below:

	length condition	excluded inducing characters
$z_p = 0$	n_1, n_2 arbitrary bounded	principal
$z_p > 0$	$n_1 + n_2 + 6\kappa z_p \leq M$	Θ

The permitted ℓ_{mesh} is independent of the number of prime factors and uniform for $\kappa \in [3/4, 1]$; constants and finite seminorm orders may depend on the fixed number of factors.

In this application $\beta_* = (1 + \kappa)/2$ by (1.3), so the zero-free premise holds by equality. Cancellation in the unweighted-moment argument. Ordinary Hecke reflection first reduces the length ranges. Two finite transforms then reduce the combined range exponent M . Transformed characters in Θ have possible main terms of the full summation size. For longer inputs the source uses the difference with fixed ideals $\mathbf{b} = (b_1, b_2)$, primary ideal variables l_1, l_2 , and positive real scales X_i, Y_i :

$$(9.5) \quad D_{\mathbf{b}}(l_1, l_2) = \prod_{i=1}^2 W_i(q_{b_i} q_{l_i} / X_i) - \prod_{i=1}^2 W_i(q_{b_i} q_{l_i} / Y_i), \quad X_1 X_2 = Y_1 Y_2.$$

The two transforms preserve, on each side of the squared norm, one character, one coprimality condition and one norm twist on the full product of prime variables and unrestricted ideal variables [1, Lemma 18.2]. For an inducing character in Θ , fixing the prime variables leaves $\vartheta(l_1) \vartheta(l_2) 1_{(l_1 l_2, \mathfrak{r}_*)=1} q_{l_1 l_2}^{it} D_{\mathbf{b}}(l_1, l_2)$.

Here $\vartheta \in \Theta$, $\mathfrak{r}_*$ is a squarefree coprimality modulus with $q_{\mathfrak{r}_*} \leq Z^{B_{\text{cop}}}$ for fixed B_{cop} , and $t \in \mathbb{R}$ is the common norm-twist height. The W_i in this argument are the two annular ideal weights, distinct from the earlier prime-interval weights. By [1, Lemma 18.3], the two product main terms have a common coefficient, independent of X_i, Y_i , multiplying respectively $(X_1 X_2)^{1+it}$ and $(Y_1 Y_2)^{1+it}$. Hence

$$(X_1 X_2)^{1+it} - (Y_1 Y_2)^{1+it} = 0,$$

which is the decisive cancellation in (9.5). The cited lemma supplies the uniform Poisson remainder. Direct volume bounds suffice first in the uncentered range $n_1 + n_2 + z_p \leq 5M/6$; the cancellation handles the remaining ranges. Increasing-width induction proves all zero-prime cases before the positive-prime cases. Uniform aggregation of support errors into $O_K(1/\log Z)$ keeps the maximum-length restriction independent of K . This cancellation is what permits the recursive estimate to close without discarding those transformed characters.

10. COUNTING ELEMENTS FROM SIMULTANEOUS LARGE VALUES

10.1. Prime sum size and admissible lengths. For a fixed Mellin parameter z on those contours, put

$$Q_i(u; z) = P_i^{-1/2} \sum_{p \in P_i} \overline{\chi_p(u)} W_i(q_p / P_i) (q_p / P_i)^{z-1}, \quad \mathcal{Q}_i = -P_i^{z-1/2} Q_i.$$

The logarithmic-derivative estimate in the zero-free rectangle gives $|Q_i| \ll U^{\varepsilon_1} P_i^{\delta/2 + O(\varepsilon_b)}$ [1, Lemma 19.1]. Choose a fixed small width for the exponent intervals $\theta_{\text{bin}} > 0$. Partition by real exponents g_i so

$$|Q_i| \leq P_i^{g_i + \theta_{\text{bin}}}, \quad g_i > 0 \implies |Q_i| \geq P_i^{g_i}, \quad 0 \leq g_i \leq \delta/2.$$

For an error factor set $g_i = 0$. Write

$$(10.1) \quad \bar{g} = \frac{\sum_i \ell_i g_i}{\ell}, \quad 0 \leq \bar{g} \leq \delta/2, \quad x = \bar{g}/\delta \in [0, 1/2].$$

Ordering positive exponents decreasingly and selecting an initial set of prime sums of total length at most z_p gives

$$(10.2) \quad \left| \prod_{i \text{ selected}} Q_i \right|^2 \geq U^{2\bar{g}z_p - \delta \max_i(\ell_i/d)},$$

up to a fixed small length decrement. Fractional selection has gain at least $\bar{g}z_p$; deleting its last partial interval gives the displayed error.

After selecting a presentation, take $\psi(n) = \nu(n) \overline{\chi_n(u)}$. The prime coefficient is $\overline{\nu(p)} 1_{p \in 1_{\mathcal{T}}}$, a fixed finite combination of Θ -characters. Conjugate the whole unweighted product when using (9.4). An element with a good prime divisor induces outside Θ . The required moment hypotheses therefore hold; smooth parameter estimates handle the allowed element-dependent profiles at fixed polynomial height cost.

Measure lengths in base U : the variable u has norm $\asymp U$, so its range exponent is one. The largest total prime lengths allowed by the two moment inequalities are

$$(10.3) \quad z_M(r) = \frac{1-r}{2}, \quad z_S(m) = \frac{1-2m}{6\kappa} = \frac{1-2m}{9/2 + 12\Delta}.$$

Using (5.5), (10.2) and the moments,

$$(10.4) \quad \#\mathcal{B} \ll \min \{ U^{1-\delta r - 2\bar{g}z_M + \varepsilon}, U^{1-2\delta m - 2\bar{g}z_S + \varepsilon} \} (1 + T_1)^A$$

where the respective lengths are admissible. The second bound uses two copies of S_m , hence $|S_m|^4$.

10.2. Optimization of the two estimates.

Proposition 10.1 (Refined character count, [1, Proposition 19.2]). *Assume the Stage-2 setting (1.3), so $\Delta = \beta_* - 7/8 \in (0, 1/24]$ and $\kappa = 3/4 + 2\Delta$. Let $\mathcal{B}$ be a fixed zero class and prime-sum size class, with $q_u \asymp U = Z^d$, $a > a_0$, $0 < \delta \leq \alpha$, $0 \leq \bar{g} \leq \delta/2$ and $x = \bar{g}/\delta$. Assume the detector height hypotheses and that the available prime length ℓ/d exceeds $7/37$ by a fixed positive amount. Put*

$$D_x = 3 - 17x/9, \quad P_x = (2 - 8x/9)(1 - x).$$

For $t \in [1, 3/2]$ define

$$(10.5) \quad R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t).$$

$$(10.6) \quad R_{\text{long}}(t) = 1 - \delta + (\alpha - \delta)(t - 1), \quad \alpha = 5/6.$$

Then, for every $\varepsilon > 0$, the length decrements and maximum individual prime length can be chosen so that

$$(10.7) \quad \#\mathcal{B} \ll U^{\max(R_{\text{short}}(t), R_{\text{long}}(t)) + \Delta/4 + \varepsilon} (1 + T_1)^A.$$

The constants and exponent A are uniform in u ; their arithmetic dependence is on the fixed data. With no selected prime sums, $t = 1$ instead gives $\#\mathcal{B} \ll U^{1-2\delta/3+\varepsilon} (1 + T_1)^A$ without any hypothesis on the available prime length.

The character-count exponent calculation. At $\delta = \alpha$, choose the detector parameter $t = 3/2$, so $r \geq 1 - O(\varepsilon)$. Sixth-power amplification then gives $\#\mathcal{B} \ll U^{1-\delta+\varepsilon}(1+T_1)^A$, which is at least as strong as (10.7); hence assume $\delta < \alpha$. For short inverse length $r < 1$ and $m < 1/2$, compare the unweighted estimate first with its $\Delta = 0$ prime length. Using $m \geq t - r - O(\varepsilon)$ gives

$$A_I(r) = 1 - \delta\{x + (1-x)r\}, \quad A_S(r) = 1 - \delta\{4x/9 + (2-8x/9)(t-r)\}.$$

Let

$$r_*(t) = \frac{(2-8x/9)t - 5x/9}{D_x}.$$

Then $r_*(t) \geq 23/37$ and $1/3 \leq t - r_*(t) \leq 1/2$. The optimal common upper exponent is (10.5).

On the inverse side a fixed decrement $\nu_0 > 0$ ensures

$$1 - r - 2z_p \geq 2\nu_0, \quad 3 - 2r - 8z_p = 4(1 - r - 2z_p) + (2r - 1), \quad 2r - 1 \geq 9/37.$$

Its required prime length is at most $7/37$, whereas $\ell/h = 8/39 > 7/37$. On the unweighted side, restoring the actual length in (10.3) costs at most

$$\begin{aligned} 2\bar{g}(1-2m) \left(\frac{2}{9} - \frac{1}{9/2+12\Delta} \right) &\leq \frac{5}{6} \left(\frac{1}{3} + O(\varepsilon) \right) \frac{16}{27} \Delta \\ &\leq \frac{40}{243} \Delta + O(\varepsilon) < \Delta/4 + O(\varepsilon). \end{aligned}$$

Here $\bar{g} \leq \delta/2 \leq 5/12$, $m \geq t - r_*(t) - O(\varepsilon) \geq 1/3 - O(\varepsilon)$, and $\frac{2}{9} - \frac{1}{9/2+12\Delta} \leq \frac{16}{27} \Delta$. For $r \geq 1$, (9.3) instead gives (10.6).

If an admissible prime length is zero or very small, the unweighted moments are used. These comparisons give the count exponent in (10.7). With no selected prime sums and $t = 1$, balance $1 - \delta r$ and $1 - 2\delta(1 - r)$ at $r = 2/3$ to obtain the stated unweighted count.

11. CONTOUR ESTIMATES AND THE ENDPOINT INEQUALITY

11.1. The exponent of a nonprincipal contribution. Move the original zero-class integral with the holomorphic correction (8.2) to $s_r = a + 16\epsilon_b$, $w_r = 1 - a - 6\epsilon_b$, $z_r = z_0 := 17/50$. Only afterward subdivide pointwise by prime-sum sizes. For a subset with cardinality $\ll U^{R+\varepsilon}(1+T_1)^A$, the factors, before small losses, are

$$\underbrace{X^{1/2-z_0} Z^{a+z_0-1} Y^{-a}}_{\text{Mellin powers}} \underbrace{U^{R+\delta/2-z_0}}_{\text{count and numerator}} \underbrace{Z^{\ell(z_0-1/2)+\bar{g}\ell}}_{\text{prime sums}}.$$

The numerator and all error factors are estimated jointly by (8.3). Subtracting $C(7/8) = 3/16$ gives

$$\begin{aligned} (11.1) \quad E(d) &= a - \frac{7}{8} + h(z_0 - 1/6) - al_y - \ell/2 + \bar{g}\ell + d(R + \delta/2 - z_0) \\ &= -\frac{1}{48} + \frac{2}{3}\delta + \frac{\bar{g}}{6} - h(1 - R) + (d - h)(R + \delta/2 - z_0) \quad [1, \text{Lemma 20.1}]. \end{aligned}$$

The integral is bounded by the supremum of this exponent over subsets defined by the sizes of the prime sums at each fixed Mellin parameter. The contour shift is made before this subdivision, so no holomorphic continuation of a subset is required.

11.2. Choice of the truncation exponent and positivity.

Lemma 11.1 (Endpoint saving, [1, Lemma 20.2]). *Let $1/50 < \delta < \alpha = 5/6$ and $0 \leq x \leq 1/2$, with D_x, P_x as above and the fixed scales (7.1). Choose*

$$(11.2) \quad \mathcal{J} = (\alpha - \delta)D_x + \delta P_x, \quad t = 1 + \frac{\delta P_x}{2\mathcal{J}}, \quad R_* = R_{\text{short}}(t) = R_{\text{long}}(t).$$

Let $E_ = E(h)$ in (11.1) with $\bar{g} = x\delta$ and $R = R_*$. Then $-E_* \geq 49/440640$. If the count loss $\Delta/4$ is included and all remaining exponent losses have sum $\varepsilon_{\text{total}}$, then*

$$(11.3) \quad E_{\text{actual}}(h) - \Delta \leq -\frac{49}{440640} - \frac{51}{64}\Delta + \varepsilon_{\text{total}}.$$

Here E_{actual} denotes the resulting exponent relative to $C(7/8)$.

The endpoint exponent calculation. Here $1 < t < 3/2$ and $35/54 \leq \mathcal{J} \leq 5/2$. Put

$$\begin{aligned} y &= 1/2 - x \in [0, 1/2], \quad v_y = 51 + 41y, \quad p_y = 7 + 18y + 8y^2, \\ j_y &= 185 + 170y + (-138 + 12y + 96y^2)\delta. \end{aligned}$$

Then $P_x = p_y/9$ and $\mathcal{J} = j_y/108$. The source's polynomial identity is

$$(11.4) \quad \begin{aligned} 10368v_y\mathcal{J}(-E_*) &= (3 + 5y)\{(4v_y\delta - 79)^2 + 49\} \\ &\quad + 4y\{4v_y\delta[(1 + 3y)(15 + 32y)\delta + 9 - 13y] + 265 + 3485y\}. \end{aligned}$$

Since $9 - 13y \geq 5/2$ and $v_y \leq 17(3 + 5y)$,

$$(11.5) \quad -E_* \geq \frac{49}{176256\mathcal{J}} \geq \frac{49}{440640} > \frac{1}{10000} \quad [1, \text{Lemma 20.2}].$$

Restoring the $\Delta/4$ in (10.7), the coefficient of R in (11.1) is h . Since $C(\beta_*) - C(7/8) = \Delta$, this gives (11.3). Here E_{actual} includes the $\Delta/4$ count loss, and $\varepsilon_{\text{total}}$ denotes the sum of the small contour, length, size-partition and smoothing losses. The subtraction of Δ is essential.

At $\delta = \alpha$, use $t = 3/2$ and unweighted amplification: $R = 1 - \delta$, whence $E(h) \leq -1/48 - \delta/16$. At $a = a_0$ use the direct count $R = 1$, giving $E(h) \leq -7/1200$. No selected zero or vanishing strict moment margin is needed in these respective boundary cases.

11.3. All frequency ranges. Let $d_{\min} = 1/100$ and $0 < \zeta < 1/48$. The estimates covering $U = Z^d$ are:

- (1) $d < d_{\min}$: absolute tuple bounds on $(s_r, w_r, z_r) = (\beta_* + \epsilon_b, 1/2, 17/50)$ give saving $79/800 - 2d_{\min} = 63/800$ relative to $C(\beta_*)$. Here $2d_{\min}$ conservatively bounds the scale cost $63d/50$ for $d < d_{\min}$.
- (2) $d_{\min} \leq d \leq 1/2$: using $R = 76/75 - 2\delta/3$ (also valid at $a = a_0$),

$$E(d) \leq E(1/2) \leq -529/2400 + (25/96)\delta \leq -49/14400.$$

The slope in d is $R + \delta/2 - z_0 = 101/150 - \delta/6 > 0$, since $\delta \leq 5/6$; this justifies $E(d) \leq E(1/2)$.

- (3) $1/2 \leq d \leq h$: the total available prime length is $\ell/d \geq 8/39 > 7/37$, and $R + \delta/2 - z_0 \geq 33/50 - \delta/2 \geq 4/25 > 0$. Thus (11.3) controls these ranges.
- (4) $h < d \leq h + \zeta$: the total available prime length remains $> 1/5 > 7/37$. The frequency slope is below two, so the additional cost is at most 2ζ .
- (5) $d > h + \zeta$: absolute lines $(2, 2, z_\infty)$ give

$$\sum_{U > Z^{h+\zeta}} |\mathcal{R}_\eta(U; Z)| \ll Z^{53/32 + (h+\zeta)(1+\varepsilon) - \zeta z_\infty + \varepsilon}.$$

Here U runs over dyadic norm scales, and $\mathcal{R}_\eta(U; Z)$ denotes the modified nonprincipal contribution from $q_u \asymp U$ in (4.7), before normalization, on these absolute lines [1, Lemma 16.2]. Choose the fixed $z_\infty > 2$ sufficiently large.

The outer estimates are [1, Lemmas 10.6, 16.2]. Only the ranges $d \geq 1/2$ use selected prime sums in the character-count moments.

12. PARAMETER CHOICES AND THE FINAL COMPARISON

The lowest zero class and the intermediate norm range satisfy $E \leq -7/1200$ and $E \leq -49/14400$, respectively, relative to $C(7/8)$. Relative to $C(\beta_*) = C(7/8) + \Delta$, they therefore save at least Δ . Since $51\Delta/64 < \Delta$, both are already covered by m_{high} below. The common real margins are

$$m_{\text{high}} = \min\{51\Delta/64, 63/800\}, \quad m_w = 23/960, \quad m_z = 13/9600.$$

Choose small moment errors and strict length decrements. Obtain the maximum permissible individual prime length $\ell_{\text{mesh}} > 0$ from [1, Lemma 18.1] before choosing K . Then choose a fixed even K with equal $\ell_i = \ell/K$ and $2\ell/K < \ell_{\text{mesh}}$ and smaller than the allowed exponent loss from discarding a partial prime interval. Smooth weights in disjoint subintervals of $(1, 2)$ give disjoint prime intervals at the common scale $Z^{\ell/K}$. Thus $\ell_i/d \leq 2\ell/K$ throughout the selected ranges. Choose $0 < \kappa_P < (7/8)\ell/K$, then the width of the prime-sum size intervals, ϵ_b , ζ , z_∞ and P_0 within the remaining margins. All these choices precede the target.

One may retain

$$(12.1) \quad \mu_0 = \frac{1}{4} \min\{m_{\text{high}}, m_w, m_z, \kappa_P\} > 0, \quad \omega_0 = \Delta/2.$$

After fixing η and its arithmetic data, choose all internal seminorm orders. [1, Lemmas 10.2–10.4] and the preceding estimates give

$$(12.2) \quad |J_\eta - f_\eta| \ll_{\eta, N} Z^{C(\beta_*) - \mu_0} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}.$$

A_η, B_η are finite nonnegative exponents fixed before the external decay order $N \in \mathbb{Z}_{\geq 0}$. The later N_η is a sufficiently large choice of this order. There is a fixed $\tau_{0, \eta} > 0$ ensuring all zero-detection height conditions for $T_1 = Z^\tau$, $0 < \tau \leq \min(d_{\min}/100, \tau_{0, \eta})$.

Choose

$$0 < \tau_\eta \leq \min\left\{d_{\min}/100, \tau_{0, \eta}, \frac{\mu_0}{4(A_\eta + 1)}\right\}, \quad B_\eta - N_\eta \tau_\eta < C(\beta_*) - \mu_0/2.$$

The first term in (12.2) saves at least $3\mu_0/4$, and the second saves $\mu_0/2$. Increasing external derivative orders changes only seminorm constants and thresholds. This is the height choice of [1, Lemma 11.1]; the full order is [1, Proposition 20.3]. Together with (8.6),

$$(12.3) \quad |J_\eta(Z)| \ll_\eta Z^{3/16 + \Delta/2}, \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*) - \mu_0/2}.$$

The exponents are independent of η . Neither J_η nor f_η contains T_1 .

13. MELLIN CONTINUATION AND CONTRADICTION

Theorem 13.1 (Hecke conclusion, [1, Theorem 1.1]). *The family supremum in (1.2) satisfies $\beta_* \leq 7/8$. Consequently every finite-order Hecke L -function over F has no zeros in $\text{Re } s > 7/8$, with principal poles allowed.*

The final continuation contradiction. Assume $\beta_* > 7/8$. Apply Proposition 3.1 with

$$\sigma_0 = 7/8, \quad c = -11/16, \quad \omega_0 = \Delta/2, \quad \sigma = \mu_0/2.$$

Holomorphy of H_η follows from (4.9) and (4.10); the chosen cutoff gives (4.11). Together with (12.3), these verify the criterion's hypotheses; $C(7/8) = 3/16$ and $0 < \omega_0 < \beta_* - 7/8$. Its contradiction therefore gives

$$(13.1) \quad \boxed{\beta_* \leq 7/8}.$$

For an imprimitive character η induced by η^* ,

$$L_F(s, \eta) = L_F(s, \eta^*) \prod_{\substack{\mathfrak{p} \mid \mathfrak{f}_\eta \\ \mathfrak{p} \nmid \mathfrak{f}_{\eta^*}}} (1 - \eta^*(\mathfrak{p})q_{\mathfrak{p}}^{-s}).$$

Here $\mathfrak{f}_\eta$ is the defining modulus and $\mathfrak{f}_{\eta^*}$ the primitive conductor. Each finite factor is nonzero for $\operatorname{Re} s > 0$, giving the same conclusion for every presentation.

14. TRANSFER TO DIRICHLET L -FUNCTIONS

Corollary 14.1 (Dirichlet conclusion, [1, Proposition 11.3]). *Every Dirichlet L -function, including ζ , has no zeros in $\operatorname{Re} s > 7/8$, with principal poles at $s = 1$ allowed.*

The Dirichlet factorization. For a primitive Dirichlet character χ of conductor q , define $\eta(\mathfrak{a}) = \chi(N\mathfrak{a})$ on ideals coprime to $3q$, and let χ_{-3} be the primitive quadratic Dirichlet character associated with $F/\mathbb{Q}$, given by the Kronecker symbol $\chi_{-3}(n) = (-3/n)$. Deleting primes dividing $3q$ and their prime ideals,

$$(14.1) \quad L_F^{S_F}(s, \eta) = L^{S_{\mathbb{Q}}}(s, \chi) L^{S_{\mathbb{Q}}}(s, \chi\chi_{-3}) \quad [1, \text{Proposition 11.3}].$$

The split and inert Euler factors give (14.1) in absolute convergence; meromorphic continuation extends the identity [1, Proposition 11.3]. Both Dirichlet factors are holomorphic on $7/8 < \operatorname{Re} s \leq 1$, $s \neq 1$. A zero of either is therefore a zero of their product, contradicting the Hecke conclusion. Euler convergence handles $\operatorname{Re} s > 1$. At $s = 1$ a possible principal pole pairs only with χ_{-3} , whose value is

$$L(1, \chi_{-3}) = \int_0^1 \frac{dx}{1+x+x^2} = \frac{\pi}{3\sqrt{3}} > 0.$$

Finally, finite Euler deletions transfer the result to all imprimitive Dirichlet presentations. The character modulo one gives ζ .

Part II. A rational extension of the bound

15. STATEMENT AND CHOICE OF PARAMETERS

This part first establishes the common estimates used by both extensions, then proves a rational improvement with an explicit calculation in the Bernstein basis. We vary the lengths in the modified average of Section 7; all arithmetic conventions, zero extensions, smooth tests and character families remain those of Part I. Throughout Parts II and III we use the proved intermediate results of [1]. In particular, we apply Theorem 13.1 before making the improved comparison.

Theorem 15.1 (Rational extension). *Every finite-order Hecke L -function over F is zero-free in*

$$\operatorname{Re} s > B_r, \quad B_r = 34999/40000.$$

The same assertion holds for every Dirichlet L -function. Principal poles are allowed, and the boundary line is excluded.

For clarity, the parameters that change are displayed together:

$$\begin{aligned}
 \vartheta_r &= \frac{1}{10000}, \quad b_r = \frac{1}{8}, \quad \ell_r = \frac{1}{6} + \vartheta_r = \frac{5003}{30000}, \\
 b &= b_r, \quad \ell = \ell_r, \\
 (15.1) \quad l_x &= \frac{1-b-\ell}{2} = \frac{21247}{60000}, \quad l_y = \frac{1+b-\ell}{2} = \frac{28747}{60000}, \quad h = 1 - l_x + \ell = \frac{16253}{20000}, \\
 X &= Z^{l_x}, \quad Y = Z^{l_y}, \quad P_i = Z^{\ell_i}, \quad \sum_{i=1}^K \ell_i = \ell.
 \end{aligned}$$

The pair (b_r, ℓ_r) specifies the rational extension. We use $b = b_r$ and $\ell = \ell_r$ at this geometry, and retain b, ℓ for the general parameter formulas shared with Part III. Here $b = l_y - l_x$ and K is a fixed even integer. Its disjoint prime sets $\mathcal{P}_i$ lie in the class $1_{\mathcal{T}}$ and outside S , as in Part I. The bin width ϵ_b is independent of ϑ_r . We retain

$$a_0 = 51/100, \quad \delta = 2a - 1, \quad x = \bar{g}/\delta \in [0, 1/2], \quad z_0 = 17/50, \quad \alpha = 5/6.$$

Thus x denotes an amplitude ratio, whereas l_x denotes a length. In all applications $\delta \geq 1/50$, so the ratio is defined. Its extension to $\delta = 0$ is used only in polynomial inequalities.

The balanced condition $l_x + l_y + \ell = 1$ gives

$$(15.2) \quad C_b(s) = s - \frac{4+b}{6}, \quad B(\ell) = \frac{11}{12} - \frac{\ell}{4}, \quad C_b(B(\ell)) = \frac{l_x}{2} + \frac{b}{12}.$$

The shared balanced hypotheses for the extensions are

$$\begin{aligned}
 l_x + l_y + \ell &= 1, \quad b = l_y - l_x > 0, \quad h = 1 - l_x + \ell, \\
 (15.3) \quad l_x &> \ell > 0, \quad l_y > \ell, \quad l_y - \ell - 11b/6 > 0, \quad 1 - 3\ell > 0, \quad \ell < 1/5, \\
 \ell/h &\geq 7/37 + \rho_{\text{sup}}, \quad \rho_{\text{sup}} > 0, \quad 5\ell > h, \\
 B &= B(\ell) = 11/12 - \ell/4 \in [437/500, 7/8].
 \end{aligned}$$

The positive reserve ρ_{sup} is fixed independently of the target. The conditions on rescaled lengths and the additive estimate are separated from the numerical choice of geometry. For (15.1), $C_b(s) = s - 11/16$ and $B(\ell) = B_r$. The hypotheses are verified by

$$\begin{aligned}
 l_x - \ell &= \frac{3}{16} - \frac{3}{2}\vartheta_r > 0, \quad l_y - \ell = \frac{5}{16} - \frac{3}{2}\vartheta_r > 0, \\
 (15.4) \quad l_y - \ell - \frac{11b}{6} &= \frac{1}{12} - \frac{3}{2}\vartheta_r > 0, \quad 1 - 3\ell > 0, \quad \ell < \frac{1}{5}, \\
 \frac{\ell}{h} &> \frac{8}{39} > \frac{7}{37}, \quad 5\ell - h = \frac{1}{48} + \frac{7}{2}\vartheta_r > 0.
 \end{aligned}$$

They ensure that the same finite modification defines an average at the new scales, that all rescaled lengths remain positive, and that there are enough prime factors for the moments with selected prime factors.

16. THE DIRECT ESTIMATE AT BALANCED SCALES

The specialized bound (7.4) does not by itself justify changing ℓ . We instead use the general reflected estimate [1, Lemma 14.3] and the additive estimate (7.5). If $\mathcal{I} \subset \{1, \dots, K\}$ is the subset of rescaled factors, put

$$d_{\mathcal{I}} = \sum_{i \in \mathcal{I}} \ell_i, \quad \ell' = \ell - d_{\mathcal{I}}, \quad M' = 1 - \ell - 2d_{\mathcal{I}}, \quad X' = Z^{l_x - d_{\mathcal{I}}}, \quad Y' = Z^{l_y - d_{\mathcal{I}}}.$$

The symbol $d_{\mathcal{I}}$ is unrelated to the norm exponent d used later. For a fixed ray class σ and rescaled prime tuple, let $B_{a,\sigma}^{\mathcal{I}}$ denote the completed coefficient $B_{m,\sigma}^J$ of (7.4), with $J = \mathcal{I}$, the row index renamed a , and the scales replaced by X', Y' above. The primes $i \notin \mathcal{I}$ are already summed, with their divisibility restrictions imposed. Put $Q = q_{b*} X' Y'$.

Lemma 16.1 (Balanced reflected estimate). *Under the coefficient and smooth-weight hypotheses of [1, Lemma 14.3], the completed row of surviving prime length ℓ' and row length M' has squared norm*

$$(16.1) \quad \sum_{q_a \ll Q} |B_{a,\sigma}^{\mathcal{I}}|^2 \ll Z^{M' + (d_{\mathcal{I}} + 5\ell - 1)_+ / 4 + \varepsilon}.$$

Here $(t)_+ = \max(t, 0)$ for a real argument t ; the loss $\varepsilon > 0$ is arbitrary.

Proof. We record the exponent calculation, retaining the normalization of (7.4). Write O for the length of the powerful part, H for the residual squarefree length, and $\Delta_H = M' - O - H$. The local auxiliary lengths in [1, Lemma 14.3] satisfy, up to losses that can be made arbitrarily small,

$$O, \Delta_H, S_0, B_0 \geq 0, \quad 2A_0 \leq O, \quad 0 \leq N_0 \leq A_0, \quad z_a \leq \ell',$$

$$T_d = 2H + 2A_0 + 2z_a - 1 - \ell' - N_0 - 3B_0, \quad v + 3\ell_b + e_\lambda \leq T_d.$$

Here A_0, N_0, B_0, S_0 describe the local divisor decompositions, z_a is their retained primal prime length, v, ℓ_b, e_λ are the retained dual lengths, and T_d is the available dual length. These variables occur only in this reflection calculation. Since $M' + \ell' - 1 = -3d_{\mathcal{I}}$,

$$T_d \leq H - 3d_{\mathcal{I}} \leq H.$$

The squared-energy exponent supplied by the cited lemma is

$$\begin{aligned} E_{\text{ref}} &= O/2 + H - S_0 - B_0 + z_a - u_{\text{hyb}} - \ell_b - 2e_\lambda/3 \\ &\quad - (T_d - v - 3\ell_b - e_\lambda)_+ / 2, \quad u_{\text{hyb}} = \min\{v, z_a, (v + z_a)/3\}. \end{aligned}$$

If $u_{\text{hyb}} = z_a$, then $E_{\text{ref}} \leq M'$. Otherwise $u_{\text{hyb}} \geq v/2$, and

$$u_{\text{hyb}} + \ell_b + 2e_\lambda/3 + (T_d - v - 3\ell_b - e_\lambda)_+ / 2 \geq T_d/4.$$

Substituting T_d , using $N_0 \leq A_0$ and $z_a \leq \ell'$, gives

$$E_{\text{ref}} \leq M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H}{4} \leq M' + \frac{d_{\mathcal{I}} + 5\ell - 1}{4}.$$

The maximum of the two bounds proves (16.1). Fixed annular offsets are $O(1/\log Z)$; the Gaussian partition, margins in the prime-length restrictions and logarithmically many dyads fit into ε . Separation is performed on the joint weight before positivity or squaring, so the separated coefficients remain independent of the row. $\square$

Proposition 16.2 (Balanced direct bound). *Assume $l_x + l_y + \ell = 1$, $b = l_y - l_x > 0$, and*

$$l_x > \ell > 0, \quad l_y > \ell, \quad l_y - \ell - 11b/6 > 0, \quad 1 - 3\ell > 0, \quad \ell < 1/5.$$

Then

$$(16.2) \quad |I_{\eta, \text{mod}}(Z)| \ll_{\eta, \varepsilon} Z^{l_x/2 + b/12 + \varepsilon}.$$

Proof. Apply the separation of (4.6) to each term of (7.2). In (7.5) the additive scale is $P_a = q_{b*}^{-1} Z^b$, and the inequality $l_y - d_{\mathcal{I}} - 11b/6 > 0$ implies $P_a^2/Y' \ll P_a^{1/6}$. Cauchy's inequality and Lemma 16.1 bound each unscaled tuple by

$$(X')^{1/2} P_a^{1/12} Z^{(d_{\mathcal{I}} + 5\ell - 1)_+ / 8 + \varepsilon}.$$

The number of rescaled tuples is $\ll Z^{d_{\mathcal{I}}+\varepsilon}$. Their coefficient is $O(Z^{-3d_{\mathcal{I}}/2})$, and replacing X' by X contributes $Z^{-d_{\mathcal{I}}/2}$. Thus their net additional exponent is

$$-d_{\mathcal{I}} + (d_{\mathcal{I}} + 5\ell - 1)_+ / 8 \leq -7d_{\mathcal{I}}/8 \leq 0.$$

Sum over the finitely many subsets. This proves the claim without invoking the specialized direct bound at altered scales. $\square$

17. EULER PRODUCTS AND EXTRACTION OF THE PRINCIPAL TERM

We establish a domain that serves both extensions. Let

$$(17.1) \quad \frac{437}{500} \leq B \leq \frac{7}{8}, \quad \mathcal{D}_2(B) = \{\operatorname{Re} s \geq B, \operatorname{Re} w \geq 19/20, \operatorname{Re} z \geq 33/200\}.$$

The first domain (4.8) remains available, with a positive margin in $\operatorname{Re}(s + w) > 1$. All Euler products use the same set S as the modified average. A fixed prime cutoff is chosen first; the target's conductor primes may then be added to S .

Lemma 17.1 (Extended principal domain). *The unselected and selected local expressions of Part I are holomorphic on $\mathcal{D}_2(B)$. Their full correction (8.2) is defined there by the same local holomorphic expressions. For sufficiently large fixed cutoff P_0 , the principal correction satisfies*

$$(17.2) \quad H_{\eta}(s) = 1 + O(P_0^{-4/5}), \quad \sup_{\operatorname{Re} s \geq B} |H_{\eta}(s) - 1| \leq \frac{1}{2}.$$

At the principal residue,

$$(17.3) \quad G_p(s, 1, 1/6)/H_p(s, 1, 1/6) = -1 + O(q_p^{-B}).$$

The constants are uniform in imaginary parts and in the unit phases.

Proof. To check the changed boundary, we give the exact local formulas from [1, Lemma 7.1]. At a good prime put $Q = q_p$, $j = v_p(u) \in \{0, \dots, 5\}$, $\rho_p = \chi_p(u/p^j)$, and

$$a_p = \overline{\varpi(p)}^3 \eta(p)^3, \quad b_p = \overline{\varpi(p)}^2 \eta(p)^2 \overline{\chi_p(4)}, \quad b_p^3 = a_p^2.$$

Here ϖ is the phase defined in Part I; these a_p, b_p are local unit phases, unrelated to a, b above. Set

$$V = Q^{-6z}, \quad R = a_p^2 Q^{4-6s-6z}, \quad W = \chi_p(u) Q^{-w}, \quad D = \eta(p) \overline{\chi_p(u)} Q^{-s}, \quad W_{\text{loc}} = \rho_p Q^{-w}.$$

At $p \mid u$, $W = D = 0$ by zero extension. With $\mathbf{1}$ denoting an indicator, the complete local formulas are

$$P_p^* = \frac{1}{1-R} \left\{ \frac{R(1-Q^{-1}) - \eta(p)(Q-1)Q^{-s-w}V^{\mathbf{1}_{j \leq 1}}}{1-V} + J_j \right\},$$

$$P_p = \frac{1}{1-V} + \mathbf{1}_{j=0} \frac{W_{\text{loc}}}{1-W_{\text{loc}}} + P_p^*, \quad H_p = P_p \frac{(1-V)(1-W)}{1-D},$$

where

j	J_j
0	$-\eta(p)\rho_p^{-1}Q^{-s} + W_{\text{loc}}R$
1	$\eta(p)Q^{-s-w}$
2	$a_p\rho_p^{-3}Q^{3/2-3s}$
3	$-\eta(p)b_p\rho_p^{-2}Q^{2-3s-w} + b_p^2\rho_p^{-4}Q^{2-4s}$
4	$-\eta(p)a_p\rho_p^{-3}Q^{5/2-4s-w}$
5	$-a_p^2Q^{3-6s}.$

Writing $\mathcal{E}_p = P_p^* + D$, cancellation gives

$$(17.4) \quad H_p - 1 = \frac{D(V + W - VW) - VW + (1 - V)(1 - W)\mathcal{E}_p}{1 - D}.$$

For $p \nmid u$, the possible exponents are

$$-s_r - w_r, \quad -s_r - 6z_r, \quad -w_r - 6z_r, \quad 4 - 6s_r - 6z_r, \quad 1 - s_r - w_r - 6z_r,$$

where $s_r = \operatorname{Re} s$, and similarly for w_r, z_r . Their maximum on (17.1) is $-B - 47/50 < -9/5$. For $p \mid u$, the table instead gives $H_p - 1 = O(Q^{1/20-B})$. The denominators $1 - R, 1 - V, 1 - D$ are uniformly separated from zero; (17.4) has no denominator $1 - W$. Normal convergence off u and a finite-product q_u^ε bound on its prime divisors follow. For $u = 1$, the tail saves $B - 3/50 > 4/5$, proving (17.2).

The selected factor is defined without dividing by H_p :

$$(17.5) \quad G_p = \frac{(\overline{\eta(p)}Q^s - Q^{-w})(1 - V)(1 - W)P_p^* - Q^{-w}(1 - VW)}{1 - D}.$$

It agrees with (8.1) on absolute lines, hence continues the finite modification by (8.2). For $u = 1$ the unselected factors are close to one, so division is permitted on the residue rectangle. The four error exponents are

$$-s_r, \quad -6z_r, \quad 4 - 5s_r - 6z_r, \quad 1 - w_r - 6z_r,$$

all at most $-B$ for the range (17.1). This proves (17.3). No nonvanishing of a nonprincipal local correction is needed. $\square$

Define $A_{\mathcal{T}}(Z)$ and c_S by (8.5), with the present ℓ_i and the same radial Mellin transform $\mathcal{M}$. The prime asymptotic (8.4) gives, for even K ,

$$A_{\mathcal{T}}(Z) > 0 \quad (Z \text{ sufficiently large}), \quad |A_{\mathcal{T}}(Z)| \asymp (\log Z)^{-K}.$$

The poles remain $w = 1$ and $z = 1/6$; the latter is independent of ℓ . For arbitrary positive l_x, l_y, ℓ , put

$$(17.6) \quad \begin{aligned} h &= 1 - l_x + \ell, & C(s) &= s + l_x/2 - 1 + h/6, \\ J_\eta(Z) &= \frac{I_{\eta, \text{mod}}(Z)}{c_S A_{\mathcal{T}}(Z)}, & f_\eta(Z) &= \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds. \end{aligned}$$

These are the same constructions as (8.8), with the changed scales. Neither function depends on a later height truncation.

Proposition 17.2 (Principal extraction). *Assume (15.3) and $\beta_* \geq B$. The individual prime lengths satisfy $\ell_i > 0$ and $\sum_{i=1}^K \ell_i = \ell$, as in (15.1). For every $0 < \kappa_P < B \min_i \ell_i$, the principal term $\mathcal{P}_\eta$ satisfies*

$$(17.7) \quad \left| \frac{\mathcal{P}_\eta(Z)}{c_S A_{\mathcal{T}}(Z)} - f_\eta(Z) \right| \ll Z^{C(\beta_*) + (1+h)\epsilon_b - l_y/20 + \varepsilon} + Z^{C(\beta_*) + \epsilon_b - h/600 + \varepsilon} + Z^{C(\beta_*) + \epsilon_b - \kappa_P + \varepsilon}.$$

Proof. The principal-row identification in Section 4 and [1, Lemmas 10.2–10.5] is arithmetic and unchanged: the only principal numerator row is $u = 1$. On that row the scalar quotient is $\zeta_F^S(6z)\zeta_F^S(w)/L_F^S(s, \eta)$. Start with $\operatorname{Re} s = \beta_* + \epsilon_b$, $\operatorname{Re} w = 1 + \epsilon_b$, $\operatorname{Re} z = 1/6 + \epsilon_b$. Move w to $19/20$, then in its residue move z to $33/200$. Lemma 17.1 justifies both moves. The raw exponent is

$$l_x/2 + \operatorname{Re} s - 1 + h \operatorname{Re} z + l_y(\operatorname{Re} w - 1).$$

The two unresidued integrals therefore save $l_y/20$ and $h/600$. At the double residue, (17.3) and the nonnegative prime tests give

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A_T(Z) (1 + O(Z^{-\kappa_P})).$$

The scalar residue is exactly c_S , including the factor $1/6$ from $\zeta_F^S(6z)$. Bound the error on its original global s line; move only the main Gaussian integral right to $\operatorname{Re} s = 2$. The reciprocal has no poles to the right of β_* , and principal L -poles become reciprocal zeros. Gaussian decay removes horizontal sides. This proves (17.7) without moving an approximation error through possible target zeros. $\square$

18. THE FIXED MOMENT PARAMETER AND THE NONPRINCIPAL ESTIMATE

Theorem 13.1 permits the fixed choice

$$(18.1) \quad \kappa = 3/4, \quad \beta_* \leq (1 + \kappa)/2.$$

It is important to fix this value rather than substitute $\kappa = 2\beta_* - 1$ below the allowed moment range. All moments below have exactly the common-character, coefficient and smooth-weight hypotheses of Section 9.

Proposition 18.1 (Fixed-parameter count). *Let $U = Z^d$ be a dyad of original sixth-power-free element rows in one zero and amplitude class. Suppose $0 < \delta \leq \alpha$, $0 \leq \bar{g} \leq \delta/2$, and $\ell/d \geq 7/37 + \rho_{\sup}$ for a fixed $\rho_{\sup} > 0$. With the D_x, P_x of Part I set*

$$(18.2) \quad \mathcal{J} = (\alpha - \delta)D_x + \delta P_x, \quad t_* = 1 + \frac{\delta P_x}{2\mathcal{J}}, \quad R_* = 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2\mathcal{J}}.$$

For every requested loss $\varepsilon > 0$, sufficiently fine fixed prime lengths give

$$(18.3) \quad \#\mathcal{B} \ll U^{R_* + \varepsilon} (1 + T_1)^A.$$

Without an assumption on the available prime length, the count without selected prime factors is $\#\mathcal{B} \ll U^{1-2\delta/3+\varepsilon} (1 + T_1)^A$.

Proof. The simultaneously large detecting polynomials of (5.5) have lengths U^r, U^m , $r + m \geq t - O(\varepsilon)$, and satisfy

$$|M_r|^2 \gg U^{\delta r - O(\varepsilon)}, \quad |S_m|^4 \gg U^{2\delta m - O(\varepsilon)}.$$

Here M_r, S_m are the Möbius-weighted and unweighted detecting polynomials defined in Section 5, with the same character and retained presentation. For total selected length U^z , the inverse moment with selected prime factors (9.1) and the unweighted moment (9.4) give the same set two count bounds:

$$U^{1-\delta r - 2\bar{g}z + \varepsilon} (1 + T_1)^A, \quad U^{1-2\delta m - 2\bar{g}z + \varepsilon} (1 + T_1)^A.$$

At norm scale U , the upper bounds on the total selected prime length are

$$z_M = (1 - r)/2, \quad z_S = 2(1 - 2m)/9.$$

Weighted greedy selection achieves gain at least $\bar{g}z$; rounding one fractional prime factor loses at most $\delta \max_i(\ell_i/d)$ in the squared lower bound. Decrease the selected prime-length bound by a fixed $\nu > 0$ before applying a moment; when the remaining prime-length bound is nonpositive, use the version without selected prime factors. These errors fit the requested ε .

For $r < 1, m < 1/2$, the affine count bounds without an additional prime-length restriction are

$$A_I(r) = 1 - \delta\{x + (1 - x)r\}, \quad S_t(r) = 1 - \delta\{4x/9 + (2 - 8x/9)(t - r)\}.$$

They cross at

$$r_*(t) = \frac{(2 - 8x/9)t - 5x/9}{D_x}, \quad r_*(1) \geq 23/37, \quad r_*(3/2) = 1.$$

Use the inverse estimate on $r \geq r_*$ and the unweighted estimate on $r \leq r_*$. The prime-length bound for the inverse moment is at most $7/37$, the bound for the unweighted moment is at most $2/27 + O(\varepsilon)$; the hypothesis on the available prime length suffices. On the inverse side the second strict margin is

$$3 - 2r - 8z = 4(1 - r - 2z) + (2r - 1) \geq 9/37,$$

while the first margin is $1 - r - 2z \geq 2\nu$. For short detecting polynomials, the count exponent is consequently

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t).$$

For $r \geq 1$, use the amplification estimate without selected prime factors (9.3):

$$\mathfrak{a}(r) - \delta r \leq R_{\text{long}}(t) = 1 - \delta + (\alpha - \delta)(t - 1).$$

For $m \geq 1/2$ the unweighted bound without selected prime factors is at most $1 - \delta$. Balancing the short and long bounds gives (18.2). Since $D_x - P_x = 1 + x - 8x^2/9 > 0$,

$$\mathcal{J} \geq 35/54, \quad 1 \leq t_* \leq 3/2.$$

At $\delta = \alpha$, the balanced values are $t_* = 3/2$ and $R_* = 1 - \alpha = 1/6$. At $r = 1$ the inverse prime-length bound is $z_M = 0$; use the estimate without selected prime factors (9.3). For $r \geq 1$, $\mathfrak{a}(r) - \alpha r = 1/6$. Thus the endpoint uses a moment without selected prime factors, rather than a limit of moments with selected prime factors and vanishing reserve.

All retained presentations use the same detecting polynomials in both moment estimates; there are only logarithmically many dyadic presentations. Common smooth-weight separation puts height dependence into $(1 + T_1)^A$. There is no $\Delta/4$ term from (10.7): that term compares a varying κ with $3/4$, whereas (18.1) is already exact. The common character can be written as $\psi(n) = \nu(n)\chi_n(u)$, and each selected prime coefficient is $\bar{\nu}(p)1_{p \in 1_T}$; it is a fixed finite combination of Θ -characters. Conjugate the whole unweighted product to match (9.4). Rows with a good prime divisor induce outside Θ , as in Section 10; bounded unit rows belong to the small-row estimate below. With no selected factors, $t = 1$ and the crossing $r = 2/3$ give the last assertion. $\square$

Proposition 18.2 (Exponent at arbitrary scales). *Let $B \leq \beta_*$, and assume (17.1), positive lengths and the unchanged joint-error and moment hypotheses of Part I. For a class with row exponent d , count exponent R , zero-class parameter a and amplitude mean $\bar{g}$, the normalized nonprincipal contribution is, for each prescribed loss $\varepsilon > 0$ and sufficiently small fixed contour and bin widths, bounded by*

$$Z^{C(B)+E_B(d)+\varepsilon}(1 + T_1)^A = Z^{C(\beta_*)+E_B(d)-(\beta_*-B)+\varepsilon}(1 + T_1)^A,$$

where

$$(18.4) \quad E_B(d) = a - B + h(z_0 - 1/6) - al_y - \ell/2 + \bar{g}\ell + d(R + \delta/2 - z_0), \quad z_0 = 17/50.$$

Thus $E_B(d)$ is measured relative to $C(B)$; relative to $C(\beta_*)$, the exponent is $E_B(d) - (\beta_* - B)$. On balanced scales with $B = B(\ell)$,

$$(18.5) \quad E_B(h) = -\frac{1}{4} + \frac{b}{6} + \frac{5\ell}{4} + (1/2 + \ell)\delta + \ell\bar{g} - h(1 - R).$$

Proof. Use the exact Poisson identity (4.7), whose scale factors are valid at arbitrary positive lengths. Move the full holomorphic correction before partitioning it into pointwise amplitude classes. The contour with a positive margin is

$$(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (a + 16\epsilon_b, 1 - a - 6\epsilon_b, z_0).$$

For example, first retain the global s line and move w so that $\operatorname{Re}(s + w) \geq 1 + 14\epsilon_b$; after external height truncation move s to the displayed line, where $\operatorname{Re}(s + w) = 1 + 10\epsilon_b$. The same zero-free separation and first Euler domain used in Part I apply. The joint bound (8.3) consumes the conductor deficit once for the entire product and yields

$$\sum_{u \in \mathcal{B}} |L_F^S(w, \chi_\bullet(u)) \mathfrak{H}_{\eta, u, Z}| \ll U^{R+\delta/2+\varepsilon} Z^{\ell(z_0-1/2)+\bar{g}\ell+\varepsilon} (1 + T_1)^A.$$

The remaining factors contribute

$$Z^{l_x(1/2-z_0)+a+z_0-1-al_y} U^{-z_0}.$$

Subtracting $C(B) = B + l_x/2 - 1 + h/6$ gives (18.4). Algebra using $l_x + l_y + \ell = 1$ gives (18.5). Horizontal joins and discarded heights are bounded by $Z^{B_\eta} T_1^{-N}$: the fixed internal seminorm orders are chosen before the arbitrary external integration-by-parts order N . The class sets need no holomorphic dependence on the contour variables. $\square$

19. A RATIONAL POLYNOMIAL INEQUALITY AND THE REMAINING NORM RANGES

Lemma 19.1 (Uniform rational saving). *For (15.1), $R = R_*$ and $(\delta, x) \in [0, 5/6] \times [0, 1/2]$, one has*

$$(19.1) \quad E_{B_r}(h) \leq -19/320000.$$

Proof. At the Part I geometry $\ell = 1/6, b = 1/8, h = 13/16$ denote the balanced exponent by E_0 . Put $y = 1/2 - x$ and

$$\begin{aligned} j &= 185 + 170y + (-138 + 12y + 96y^2)\delta = 108\mathcal{J}, \\ Q_0 &= 10(37 + 34y) - 8(237 + 377y + 26y^2)\delta \\ &\quad + 48(51 + 131y + 94y^2 + 32y^3)\delta^2, \quad Q = Q_0 - 96j/5000. \end{aligned}$$

Direct expansion gives

$$(19.2) \quad Q = 10368\mathcal{J}(-E_0 - 1/5000).$$

To prove its sign on the whole rectangle, use the Bernstein basis, named after Sergei Natanovich Bernstein, who introduced it in his constructive proof of Weierstrass's approximation theorem [16]; see [17] for a modern account. The Bernstein basis polynomials of degree n on $[0, 1]$ are

$$(19.3) \quad \mathbf{B}_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}, \quad 0 \leq i \leq n.$$

They satisfy

$$\mathbf{B}_{i,n}(t) \geq 0, \quad \sum_{i=0}^n \mathbf{B}_{i,n}(t) = 1 \quad (0 \leq t \leq 1).$$

On $[a_1, a_2] \times [b_1, b_2]$, substitute $\delta = a_1 + (a_2 - a_1)v$, $y = b_1 + (b_2 - b_1)w$ and write $Q = \sum_{p=0}^2 \sum_{q=0}^3 A_{pq} v^p w^q$. Its representation in the tensor-product Bernstein basis is

$$Q(\delta, y) = \sum_{i=0}^2 \sum_{k=0}^3 B_{ik} \mathbf{B}_{i,2}(v) \mathbf{B}_{k,3}(w).$$

The bidegree $(2, 3)$ Bernstein coefficients are

$$B_{ik} = \sum_{p \leq i, q \leq k} A_{pq} \frac{\binom{i}{p}}{\binom{2}{p}} \frac{\binom{k}{q}}{\binom{3}{q}}, \quad 0 \leq i \leq 2, \quad 0 \leq k \leq 3.$$

Nonnegativity and the partition-of-unity identity give

$$\min_{i,k} B_{ik} \leq Q(\delta, y) \leq \max_{i,k} B_{ik} \quad ((\delta, y) \in [a_1, a_2] \times [b_1, b_2]).$$

The listed rectangles cover the original rectangle; their complete partition and coefficients are verified by the accompanying exact arithmetic checker. The last column is a coefficient minimum, not a sampled value:

δ interval	y interval	$\min B_{ik}$
$[0, 5/24]$	$[0, 1/4]$	1380407/18000
$[5/24, 5/16]$	$[0, 1/8]$	27677/2000
$[5/16, 35/96]$	$[0, 1/16]$	12437/8000
$[35/96, 25/64]$	$[0, 1/32]$	2301/16000
$[25/64, 5/12]$	$[0, 1/32]$	12581/32000
$[35/96, 5/12]$	$[1/32, 1/16]$	2952669/1024000
$[5/16, 5/12]$	$[1/16, 1/8]$	85933/192000
$[5/24, 5/16]$	$[1/8, 1/4]$	53811/2560
$[5/16, 5/12]$	$[1/8, 1/4]$	8213/800
$[0, 5/24]$	$[1/4, 1/2]$	336467/4000
$[5/24, 5/16]$	$[1/4, 3/8]$	946273/24000
$[5/16, 5/12]$	$[1/4, 3/8]$	666871/16000
$[5/24, 5/12]$	$[3/8, 1/2]$	70651/4000
$[5/12, 5/6]$	$[0, 1/2]$	319/125

Thus $Q > 0$ throughout, so $-E_0 \geq 1/5000$. With R_*, δ, x fixed, changing ℓ at fixed b gives

$$\partial_\ell E(h) = -1/4 + \delta + \bar{g} + 3R_*/2 = 5/4 + \delta(x - 1/2) + \frac{3(\alpha - \delta)\delta P_x}{4\mathcal{J}} \leq 45/32.$$

We used $\mathcal{J} \geq \alpha P_x$ and $\delta(\alpha - \delta) \leq \alpha^2/4$. Therefore $E(h) \leq -1/5000 + 45/(32 \cdot 10000) = -19/320000$. $\square$

The endpoint inequality alone is insufficient: very short and very long rows require different contours. We record a reusable division of the norm ranges.

Lemma 19.2 (Remaining ranges). *Assume the general hypotheses (15.3), $B \leq \beta_* \leq 7/8$, and $0 \leq \delta \leq 5/6$. Then the following estimates hold, with arbitrarily small fixed losses. Set $d_{\min} = 1/100$.*

- (1) *For rows $q_u \ll Z^{d_{\min}}$, including bounded unit rows, the normalized exponent relative to $C(\beta_*)$ is at most $13h/75 - l_y/2 + 2d_{\min}$.*
- (2) *For $d_{\min} \leq d \leq 1/2$ and $\delta \geq 1/50$, use $R = 76/75 - 2\delta/3$ in (18.4). Its slope in d is $101/150 - \delta/6 > 0$; it is enough to bound $E_B(1/2)$.*
- (3) *For $1/2 \leq d \leq h$ use R_* . The slope $R_* + \delta/2 - z_0$ is at least $4/25$ and at most 2; it is enough to bound $E_B(h)$.*
- (4) *The lowest zero class $a = a_0$, without a detector, has $R = 1$ and $\bar{g} \leq 1/100$. Its endpoint exponent is at most $-1/4 + b/6 + 5\ell/4 + (1/2 + 3\ell/2)/50$.*
- (5) *Extending the third range to $h + \zeta$ costs at most 2ζ , provided $\ell/(h + \zeta) > 7/37$. Beyond $h + \zeta$, an absolute contour with sufficiently large fixed $\operatorname{Re} z = z_\infty$ gives any prescribed power saving.*

Proof. For small rows retain the global reciprocal line $\operatorname{Re} s = \beta_* + \epsilon_b$ and put $\operatorname{Re} w = 1/2$, $\operatorname{Re} z = z_0$. Use the first domain with margin $B - 1/2 > 0$. Formula (17.5) and the local table imply $G_p = O(1)$ off u , $G_p = O(q_p^{1/2})$ on u . Consequently the full correction is $\ll U^\varepsilon Z^{\ell z_0}$: off-row primes sum to $P_i^{z_0}$, and on-row primes contribute at most $U^\varepsilon P_i^{z_0-1/2}$. The nonprincipal numerator costs $U^{3/5+\varepsilon}$ on this line, by the growth estimate used in [1, Lemma 10.6]. Counting $O(U)$ original rows and the factor U^{-z_0} give $U^{63/50+\varepsilon}$. The exterior factor relative to $C(\beta_*)$ is $Z^{13h/75-l_y/2+O(\epsilon_b)}$. Since $63/50 < 2$, the first claim follows.

For intermediate rows use the count without selected prime factors in Proposition 18.1, allowing an extra $1/75$. For moderate rows use its count with selected prime factors. Differentiating (18.4) gives the slopes asserted; for the latter use $R_* \geq 1 - \delta$ and $\delta \leq 5/6$. The lowest zero class is bounded trivially by $O(U)$ and $\bar{g} \leq \delta/2$. The extension beyond h follows from the slope upper bound. Finally, on $(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (2, 2, z_\infty)$, absolute convergence gives

$$\mathcal{R}(U; Z) \ll Z^{l_x/2+1+l_y+h z_\infty+\varepsilon} U^{1+\varepsilon-z_\infty}.$$

Summing $U > Z^{h+\zeta}$ yields an exponent containing $-\zeta z_\infty$. Choose z_∞ after ζ is fixed. All statements concern the full correction, including its local zeros. $\square$

For the rational geometry the small-row saving is at least $78699/1000000$, and the saving for the lowest zero class is at least $1093/192000$. For intermediate rows, direct substitution gives

$$\max_{1/50 \leq \delta \leq 5/6} E_{B_r}(1/2) \leq -\frac{49}{14400} + \frac{177}{2000000} = -\frac{59657}{18000000}.$$

Together with Lemma 19.1, these are strict, parameter-independent margins. Choose

$$0 < \zeta < \min\{5\ell - h, (19/320000)/64\}.$$

This also ensures $\ell/(h + \zeta) > 1/5 > 7/37$.

20. COMBINING THE ESTIMATES AND THE CONTINUATION CONTRADICTION

The last step uses Proposition 3.1 rather than repeating its Mellin-continuation proof. To make the uniformity needed there explicit, we formulate the argument combining the estimates.

Proposition 20.1 (Comparison from strict exponent margins). *Assume the balanced hypotheses (15.3). Suppose also that:*

- (1) *the direct bound (16.2) with $l_x/2 + b/12 = C_b(B)$;*
- (2) *the principal estimates of Proposition 17.2;*
- (3) *the exponents for the small and intermediate norm ranges and the lowest zero class in Lemma 19.2 are negative, and $E_B(h) \leq 0$ for the count with selected prime factors.*

If $\Delta = \beta_ - B > 0$, then there is a number $\sigma > 0$, chosen before the target η , such that*

$$(20.1) \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_b(\beta_*)-\sigma}.$$

The principal factor is holomorphic on $\operatorname{Re} s > B$ and satisfies (17.2).

Proof. Replacing B by β_* in the central exponent subtracts the same Δ in every class. Choose a positive margin smaller than $\Delta/4$, the strict savings for the small and intermediate norm ranges and the lowest zero class, and $l_y/80, h/2400$. Fix ζ smaller still so that 2ζ consumes less than a quarter of that margin and all prime-length inequalities remain strict. Choose the requested moment loss, strict prime-length reserves and maximum individual prime lengths inside that margin. The hypotheses of the inverse moment with

selected prime factors hold with a strict reserve as in Proposition 18.1; the individual prime lengths satisfy the upper bound required in (9.4). Fix an even K and the actual ℓ_i before the target. Now choose $0 < \kappa_P < B \min_i \ell_i$, decrease the common margin if necessary, and finally choose ϵ_b and the remaining smooth-weight and rounding losses so that their finite sum is smaller than a quarter of this margin. All these real choices precede η .

The class count is finite apart from logarithmically many dyads and admissible polynomial configurations, whose losses are absorbed in the chosen powers. Lemmas 19.2 and 17.1, Propositions 18.1, 18.2, and 17.2 therefore give a common $\mu > 0$ with

$$(20.2) \quad |J_\eta - f_\eta| \ll_\eta Z^{C_b(\beta_*) - \mu} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}.$$

The internal smoothness orders, A_η and B_η are fixed before N . They may depend on the target; μ does not. If $\tau_{\text{det}} > 0$ is the permissible detector height exponent, choose

$$0 < \tau_\eta \leq \min\{d_{\min}/100, \tau_{\text{det}}, \mu/[4(A_\eta + 1)]\}, \quad T_1 = Z^{\tau_\eta}.$$

Then choose N so large that $B_\eta - N\tau_\eta < C_b(\beta_*) - \mu/2$. Equation (20.2) proves (20.1) with $\sigma = \mu/2$. The dependence of the analysis height on η is harmless: J_η, f_η were fixed in (17.6). $\square$

Proof of Theorem 15.1. Suppose $\Delta_r = \beta_* - B_r > 0$. Part I already gives $\beta_* \leq 7/8$, so (18.1) holds. The geometric inequalities (15.4), the direct bound, Lemma 19.1 and the remaining-range estimates verify Proposition 20.1. For any $0 < \omega_0 < \Delta_r$, the normalized direct bound is

$$|J_\eta(Z)| \ll Z^{C_b(B_r) + \omega_0}.$$

Apply Proposition 3.1 with $\sigma_0 = B_r$ and $c = -11/16$. Its principal factor is holomorphic and within $1/2$ of one by (17.2); its comparison hypothesis is (20.1). The criterion implies $\beta_* \leq B_r$, contradicting $\Delta_r > 0$. The transfer proof of Corollary 14.1 applies with this new positive boundary and gives the Dirichlet assertion. $\square$

Part III. An algebraic extension of the bound

21. ALGEBRAIC PARAMETERS AND THE SCOPE OF THE RESULT

Part II kept $b = b_r = 1/8$ fixed and improved the boundary by $1/40000$. Varying b and ℓ together improves it by $7/8 - B_{\text{new}} = 4.293020083 \dots \times 10^{-5}$. Shared analytic constructions refer to Part II; character and amplitude notation is inherited from Part I.

Theorem 21.1 (Algebraic extension). *Every finite-order Hecke L -function over F , and every Dirichlet L -function, is zero-free in*

$$(21.1) \quad \text{Re } s > B_{\text{new}} := \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799 \dots$$

Principal poles are allowed and the boundary is excluded.

Let $r_{921} = \sqrt{921}$, and set

$$(21.2) \quad \begin{aligned} \delta_c &= \frac{49 - r_{921}}{48}, & \ell_{\text{new}} &= \frac{33 + 8r_{921}}{1653}, & b_{\text{new}} &= -\frac{4}{29} + \frac{230r_{921}}{26709}, \\ l_{x,\text{new}} &= \frac{1 - b_{\text{new}} - \ell_{\text{new}}}{2}, & l_{y,\text{new}} &= \frac{1 + b_{\text{new}} - \ell_{\text{new}}}{2}, & h_{\text{new}} &= \frac{1 + b_{\text{new}} + 3\ell_{\text{new}}}{2}, \\ B_{\text{new}} &= \frac{11}{12} - \frac{\ell_{\text{new}}}{4}, & C_{\text{new}}(s) &= s - \frac{4 + b_{\text{new}}}{6}. \end{aligned}$$

The subscript new denotes the algebraic extension and its associated lengths and comparison exponent. In particular, b_{new} is a real length difference, whereas b_* in Part I is an arithmetic element. Below we write $b = b_{\text{new}}$, $\ell = \ell_{\text{new}}$, $l_x = l_{x,\text{new}}$, $l_y = l_{y,\text{new}}$ and $h = h_{\text{new}}$ when this geometry is understood. The approximate values, used only for orientation, are

$$\begin{array}{l|l} \ell_{\text{new}} & 0.166838387470\dots \\ l_{x,\text{new}} & 0.354878087771\dots \\ h_{\text{new}} & 0.811960299699\dots \end{array} \quad \begin{array}{l|l} b_{\text{new}} & 0.123405436989\dots \\ l_{y,\text{new}} & 0.478283524759\dots \\ \delta_c & 0.388583712271\dots \end{array}$$

Every sign and equality below uses the exact values in (21.2).

Theorem 21.1 is an analytic deduction from the full source inputs. Proposition 25.1 separately proves an algebraic lower bound under three explicit constraints. The expected optimality within the larger estimate system is discussed in the remark following that proposition.

22. THE DIRECT ESTIMATE WITH FREE TOTAL LENGTH

For the algebraic lower-bound calculation we allow $M = l_x + l_y$ independently of ℓ . Assume

$$l_x > \ell, \quad l_y > \ell, \quad b = l_y - l_x > 0, \quad h = 1 - l_x + \ell > 0.$$

All lengths lie in fixed bounded ranges. The signal exponent is

$$(22.1) \quad C(s) = s - 5/6 + l_x/3 + \ell/6.$$

In particular the additive constant changes when b changes; the Part I expression $s - 11/16$ cannot be retained at the algebraic scales.

Lemma 22.1 (General reflected estimate). *Under the common-weight and coefficient hypotheses of [1, Lemma 14.3], let $B_{a,\sigma}^{\mathcal{I}}$ be the completed coefficients defined in Section 16, now with row length m and surviving prime length l . Thus $X'Y' \asymp Z^m$ and $Q = q_{b_*}X'Y'$; the remaining arithmetic, coefficient and smooth-weight data are fixed as in that definition. Their squared norm satisfies*

$$(22.2) \quad \sum_{q_a \ll Q} |B_{a,\sigma}^{\mathcal{I}}|^2 \ll Z^{m+k(m,l)+\varepsilon},$$

$$k(m,l) = \max \left\{ 0, \frac{1+3l-2m}{4}, m+l-1, \frac{3l-1}{2} \right\}.$$

The estimate includes both signs of $m+l-1$.

Proof. Use the local lengths defined in the proof of Lemma 16.1, now with $H \leq m - O$, $z_a \leq l$. Suppress arbitrarily small annular and logarithmic offsets. Then

$$\begin{aligned} O, S_0, B_0, v, l_b, e_\lambda &\geq 0, \quad 2A_0 \leq O, \quad 0 \leq N_0 \leq A_0, \\ T_d &= 2H + 2A_0 + 2z_a - 1 - l - N_0 - 3B_0, \quad v + 3l_b + e_\lambda \leq T_d, \\ E_{\text{ref}} &= O/2 + \max(H, v + l_b) - S_0 - B_0 + z_a - u_{\text{hyb}} - l_b - 2e_\lambda/3 \\ &\quad - (T_d - v - 3l_b - e_\lambda)_+/2, \quad u_{\text{hyb}} = \min\{v, z_a, (v + z_a)/3\}. \end{aligned}$$

There are four exhaustive cases. If the maximum is H and $u_{\text{hyb}} = z_a$, then $E_{\text{ref}} \leq O/2 + H \leq m$. If the maximum is H and $u_{\text{hyb}} \neq z_a$, then $u_{\text{hyb}} \geq v/2$ and the dual terms save at least $T_d/4$. Substitution gives

$$\begin{aligned} E_{\text{ref}} &\leq O/2 + H/2 - A_0/2 + z_a/2 + (1 + l + N_0 + 3B_0)/4 - S_0 - B_0 \\ &\leq m/2 + (1 + 3l)/4. \end{aligned}$$

If the maximum is $v + l_b$ and $u_{\text{hyb}} = z_a$, then

$$E_{\text{ref}} \leq O/2 + v \leq O/2 + T_d \leq 2m + l - 1.$$

If the maximum is $v + l_b$ and $u_{\text{hyb}} \neq z_a$, then

$$E_{\text{ref}} \leq O/2 + v/2 + z_a \leq O/2 + T_d/2 + z_a \leq m + (3l - 1)/2.$$

Their maximum is $m + k(m, l)$.

Restore all signed $O(1/\log Z)$ offsets, annular widths and losses from restricting the summation ranges. If each is bounded by ε_0 , a common bound for losses $20\varepsilon_0$ bounds their effect in these four linear inequalities. The homogeneous joint-weight separation precedes squaring; its Gaussian translates have summable seminorms and its discarded whole annuli have arbitrary power saving. The remaining dyads are logarithmically many. Choose ε_0 inside the requested ε . This proves the bound with the source's actual coefficient classes and row normalization. $\square$

Proposition 22.2 (Full direct exponent). *The modified average satisfies*

$$(22.3) \quad |I_{\eta, \text{mod}}(Z)| \ll Z^{L_{\text{low}} + \varepsilon}, \quad L_{\text{low}} = \frac{l_x}{2} + \max\{0, b/12, b - l_y/2\} + \frac{1}{2}k(M, \ell).$$

Consequently its sufficient comparison boundary is

$$(22.4) \quad B_{\text{low}} = L_{\text{low}} + 5/6 - l_x/3 - \ell/6.$$

On balanced scales this simplifies to

$$(22.5) \quad B_{\text{low}} = \frac{11}{12} - \frac{\ell}{4} + \frac{(8b + 3\ell - 3)_+}{12} + \frac{(5\ell - 1)_+}{8}.$$

Proof. Keep all three terms in (7.5); their Cauchy contribution is $\max\{0, b/12, b - l_y/2\}$. For a rescaled subset of length $d_{\mathcal{I}}$, Lemma 22.1 applies with $m = M - 2d_{\mathcal{I}}$ and $l = \ell - d_{\mathcal{I}}$. After tuple counting and rescaling as in Proposition 16.2, its total exponent is

$$F(d_{\mathcal{I}}) = l_x/2 - d_{\mathcal{I}} + \max\{0, b/12, b - l_y/2 + d_{\mathcal{I}}/2\} + \frac{1}{2}k_{\mathcal{I}},$$

$$k_{\mathcal{I}} = \max\left\{0, \frac{1 + 3\ell - 2M + d_{\mathcal{I}}}{4}, M + \ell - 1 - 3d_{\mathcal{I}}, \frac{3\ell - 1 - 3d_{\mathcal{I}}}{2}\right\}.$$

For $d_2 > d_1$, the additive maximum increases by at most $(d_2 - d_1)/2$, and $k_{\mathcal{I}}$ by at most $(d_2 - d_1)/4$. Thus $F(d_2) - F(d_1) \leq -3(d_2 - d_1)/8$, so the empty subset controls the direct exponent. Subtract (22.1) to obtain (22.4). With $M = 1 - \ell$, the extra reflected maximum is $(5\ell - 1)_+/4$, and the extra Gram maximum is $(8b + 3\ell - 3)_+/12$, proving (22.5). $\square$

23. COMPARISON OF THE CHARACTER-COUNT ESTIMATES

The algebraic parameters have enough available prime length for Proposition 18.1. We also describe the alternatives relevant to the expected optimality of these estimates. Let $z_{\text{sup}} = \ell/d$ be the available prime length at row scale $U = Z^d$ and let $c_{\kappa} = 1/(6\kappa)$, $3/4 \leq \kappa \leq 1$. For $1 \leq t \leq 3/2$, $t - 1/2 \leq r \leq 1$ and $m = t - r$, the inverse and unweighted savings are

$$(23.1) \quad \begin{aligned} F_I(r) &= r + 2x \min\{(1 - r)/2, z_{\text{sup}}\}, \\ F_S(r) &= 2m + 2x \min\{c_{\kappa}(1 - 2m), z_{\text{sup}}\}. \end{aligned}$$

The count exponent for short detecting polynomials is $1 - \delta \min_r \max\{F_I(r), F_S(r)\}$. The two savings are increasing and decreasing respectively, and their crossing is one of

$$(23.2) \quad \begin{aligned} & \frac{2t}{3}, \quad \frac{(2 - 4c_\kappa x)t + 2c_\kappa x - 2xz_{\text{sup}}}{3 - 4c_\kappa x}, \\ & \frac{2t + 2xz_{\text{sup}} - x}{3 - x}, \quad \frac{(2 - 4c_\kappa x)t + (2c_\kappa - 1)x}{3 - x - 4c_\kappa x}. \end{aligned}$$

Each expression is used only when its corresponding capped or uncapped branches are valid. If a crossing lies outside the interval, take its appropriate endpoint. The count bound for long detecting polynomials remains $R_{\text{long}}(t)$ from Proposition 18.1.

Removing an upper bound on available prime length can only increase a saving, and increasing κ can only decrease the prime-length bound for the unweighted moment. Thus neither operation improves upon the exponent R_* obtained with no additional prime-length restriction and $\kappa = 3/4$, after optimizing t . With no selected primes, balancing the short- and long-polynomial bounds without selected prime factors instead gives

$$t_N = \frac{\alpha}{\alpha - \delta/3}, \quad R_N = 1 - \frac{2\alpha\delta}{3\alpha - \delta} \geq R_*.$$

This includes the no-prime alternative in the comparison. Strict moment reserves, coefficient restrictions and height factors are handled exactly as in Proposition 18.1. These count exponents describe the sufficient bounds furnished by the moment estimates; they are not lower bounds on actual row counts.

24. AN EXACT POLYNOMIAL PROOF OF THE ENDPOINT

At $x = 1/2$, formula (18.2) reduces to

$$(24.1) \quad R_*(\delta, 1/2) = \frac{185 - 288\delta + 96\delta^2}{185 - 138\delta}.$$

The equation $R_* = 2/3$ has smaller root δ_c , since

$$288\delta^2 - 588\delta + 185 = 0.$$

At that root the coefficient of b in (18.5) vanishes, and $E_B(h) = 0$ forces

$$\ell = \frac{5 - 6\delta_c}{9 + 18\delta_c} = \ell_{\text{new}}.$$

The stationary condition in δ gives b_{new} . These equations locate the candidate; the following identity proves its validity over the full rectangle.

Lemma 24.1 (Algebraic endpoint inequality). *At (21.2), for $(\delta, x) \in [0, 5/6] \times [0, 1/2]$, $\bar{g} = x\delta$ and $R = R_*$,*

$$(24.2) \quad E_{B_{\text{new}}}(h_{\text{new}}) \leq 0.$$

Equality occurs only at $(\delta, x) = (\delta_c, 1/2)$.

Proof. Put $y = 1/2 - x \geq 0$ and

$$p(y) = 7 + 18y + 8y^2, \quad j(y, \delta) = 185 + 170y + (-138 + 12y + 96y^2)\delta,$$

so $P_x = p(y)/9$, $\mathcal{J} = j/108$. Write $C_0 = -1/4 + b_{\text{new}}/6 + 5\ell_{\text{new}}/4$. Substitution in (18.5) gives

$$(24.3) \quad \mathcal{Q} := 216\mathcal{J}(-E_{B_{\text{new}}}(h_{\text{new}})) = -2jC_0 + j(b_{\text{new}} + 2\ell_{\text{new}}y)\delta - 12h_{\text{new}}(5/6 - \delta)\delta p(y).$$

Write $\mathcal{Q} = \mathcal{A}(y)\delta^2 + \mathcal{D}(y)\delta + \mathcal{G}(y)$; these three coefficient polynomials are defined by this identity. For $\mathcal{A}(y) = \sum_{i=0}^3 A_i y^i$, direct expansion yields

$$\begin{aligned} A_0 &= 31824/551 - 36688r_{921}/169157, \\ A_1 &= 50916/551 + 214096r_{921}/169157, \\ A_2 &= 17352/551 + 347296r_{921}/169157, \\ A_3 &= 2112/551 + 512r_{921}/551. \end{aligned}$$

Every A_i is positive. The exact discriminant identity is

$$(24.4) \quad 4\mathcal{A}(y)\mathcal{G}(y) - \mathcal{D}(y)^2 = y(d_1 + d_2y + d_3y^2 + d_4y^3),$$

where

$$\begin{aligned} d_1 &= (3861265032800 - 122715443200r_{921})/279616521, \\ d_2 &= 2386327618300/279616521 - 19093999200r_{921}/93205507, \\ d_3 &= -3389811480400/279616521 + 46823488000r_{921}/93205507, \\ d_4 &= -40221718800/4905553 + 4408611200r_{921}/14716659. \end{aligned}$$

Expansion with $r_{921}^2 = 921$ verifies the identity coefficient by coefficient; the accompanying exact arithmetic checker verifies this identity and the associated parameter inequalities. A rational isolating interval, obtained by squaring its endpoints, is

$$(24.5) \quad 30.347981810 < r_{921} < 30.347981811.$$

Substitution into these linear expressions gives

$$490 < d_1 < 491, \quad 2317 < d_2 < 2318, \quad 3122 < d_3 < 3123, \quad 892 < d_4 < 893.$$

The same bracket proves $A_0 > 0$; positivity of the other A_i is immediate. All decimal endpoints here are terminating rational numbers. Complete the square:

$$4\mathcal{A}(y)\mathcal{Q} = (2\mathcal{A}(y)\delta + \mathcal{D}(y))^2 + y(d_1 + d_2y + d_3y^2 + d_4y^3) \geq 0.$$

Since $\mathcal{J} \geq 35/54$ on the required rectangle, (24.3) proves the asserted sign. For $y > 0$ it is strict. At $y = 0$, the exact polynomial is $A_0(\delta - \delta_c)^2$, proving the equality assertion. $\square$

25. AN ALGEBRAIC LOWER BOUND FOR THE EXPONENT CONSTRAINTS

Put $M = l_x + l_y$. The two terms 0 and $M + \ell - 1$ in the reflected majorant give the direct comparison constraints

$$(25.1) \quad B \geq L_0 := 5/6 + M/12 - \ell/6, \quad B \geq L_K := 1/3 + 7M/12 + \ell/3.$$

At $(\delta, x) = (\delta_c, 1/2)$, a count exponent $R \geq 2/3$ in (18.4), with $h > 0$, also requires

$$(25.2) \quad B \geq H_c := 1 + \delta_c - (1 + \delta_c)M/2 + \delta_c\ell.$$

The following proposition is a statement about these three affine constraints. It assumes the constraints explicitly.

Proposition 25.1 (Algebraic lower bound). *Let $M, \ell, B \in \mathbb{R}$ and $\delta_c = (49 - \sqrt{921})/48$. If*

$$(25.3) \quad B \geq L_0, \quad B \geq L_K, \quad B \geq H_c,$$

with L_0, L_K, H_c defined above, then $B \geq B_{\text{new}}$. Equality is attained by $M = 1 - \ell_{\text{new}}$ and $\ell = \ell_{\text{new}}$.

Proof. The positive weights

$$w_0 = \frac{4 + 18\delta_c}{9 + 18\delta_c}, \quad w_K = \frac{2}{9 + 18\delta_c}, \quad w_H = \frac{3}{9 + 18\delta_c}$$

sum to one and satisfy

$$w_0/12 + 7w_K/12 - (1 + \delta_c)w_H/2 = 0, \quad -w_0/6 + w_K/3 + \delta_c w_H = 0.$$

Their weighted sum cancels M and ℓ , giving

$$B \geq \max\{L_0, L_K, H_c\} \geq w_0 L_0 + w_K L_K + w_H H_c = \frac{7 + 18\delta_c}{9 + 18\delta_c} = \frac{1507 - 2r_{921}}{1653} = B_{\text{new}}.$$

The cancellation proves the lower bound for every real M, ℓ ; no numerical enumeration of parameter choices is involved. At $M = 1 - \ell_{\text{new}}$ and $\ell = \ell_{\text{new}}$, direct substitution gives $L_0 = L_K = H_c = B_{\text{new}}$. $\square$

Remark on expected optimality. We expect the boundary B_{new} to be optimal for the full direct majorant of Proposition 22.2, the character-count bounds of Section 23, and the nonprincipal exponent (18.4), restricted to $3/4 \leq \kappa \leq 1$, $1 \leq t \leq 3/2$, positive scale exponents $l_x, l_y > \ell$, $b > 0$, $h > 0$, and the available-prime-length restrictions specified there. This expectation includes the option of selecting no prime factors. Its additional content is that every admissible comparison within that estimate system must satisfy the three constraints above. Proposition 25.1 proves the algebraic lower bound once those constraints hold.

26. THE AUXILIARY ESTIMATES AND THE FINAL CONTRADICTION

At (21.2), exact substitution gives

$$(26.1) \quad \begin{aligned} b_{\text{new}} > 0, \quad l_{x,\text{new}} - \ell_{\text{new}} > 0, \quad l_{y,\text{new}} - \ell_{\text{new}} > 0, \quad 1 - 3\ell_{\text{new}} > 0, \quad \ell_{\text{new}} < 1/5, \\ l_{y,\text{new}} - \ell_{\text{new}} - 11b_{\text{new}}/6 > 0, \quad \ell_{\text{new}}/h_{\text{new}} > 1/5 > 7/37, \quad 5\ell_{\text{new}} - h_{\text{new}} > 0. \end{aligned}$$

For example these inequalities follow by replacing r_{921} by the rational endpoints in (24.5) after reducing the expressions to $a + cr_{921}$; the denominators are positive. There is therefore no extra direct penalty in (22.5), and

$$(26.2) \quad |J_\eta(Z)| \ll Z^{C_{\text{new}}(B_{\text{new}}) + \varepsilon}, \quad C_{\text{new}}(B_{\text{new}}) = l_{x,\text{new}}/2 + b_{\text{new}}/12.$$

Equation (26.1) verifies the general hypotheses (15.3); the ranges in Lemma 19.2 therefore apply at these lengths. For the lowest zero class $\delta_0 = 1/50$ the saving is

$$m_F = -C_0 - (1/2 + 3\ell_{\text{new}}/2)\delta_0 > 0.$$

For intermediate rows use $R = 76/75 - 2\delta/3$; the upper endpoint exponent is bounded by

$$E_{B_{\text{new}}}(1/2) \leq \frac{(225\ell_{\text{new}} + 50 - 75b_{\text{new}})\delta + 78\ell_{\text{new}} - 49b_{\text{new}} - 73}{300} < 0 \quad (1/50 \leq \delta \leq 5/6).$$

It is affine in δ , and both endpoint values are negative by (24.5). For small rows the saving is

$$m_S = l_{y,\text{new}}/2 - 13h_{\text{new}}/75 - 2/100 > 0.$$

For moderate rows, $E_{B_{\text{new}}}(d) \leq E_{B_{\text{new}}}(h_{\text{new}}) \leq 0$ by Lemma 24.1; for $h_{\text{new}} < d \leq h_{\text{new}} + \zeta$ the cost is at most 2ζ . Very large rows are handled by the same absolute contour as in Lemma 19.2. The only non-strict central inequality is the isolated critical class $(\delta_c, 1/2)$; it is harmless under a positive contradiction gap.

The principal domain also needs no new argument:

$$437/500 < B_{\text{new}} < B_r < 7/8.$$

Lemma 17.1 and Proposition 17.2 therefore apply with $B = B_{\text{new}}$ and the new lengths. In particular, the principal factor has no zeros on $\text{Re } s > B_{\text{new}}$ and is the same factor appearing in the new signal. The first-domain small-row margin is $B_{\text{new}} - 1/2 > 0$.

The non-strict endpoint is covered by the same comparison proposition. Indeed, under the contradiction hypothesis $\Delta_{\text{new}} = \beta_* - B_{\text{new}} > 0$,

$$C_{\text{new}}(B_{\text{new}}) + E_{B_{\text{new}}}(h_{\text{new}}) - C_{\text{new}}(\beta_*) \leq -\Delta_{\text{new}}.$$

The finitely many contour, moment, binning and normalization losses have fixed finite coefficients. Proposition 20.1 chooses them inside this gap before fixing the target.

Proof of Theorem 21.1. If $\Delta_{\text{new}} > 0$, the direct bound (26.2), the extended Euler domain, the strict auxiliary estimates and the endpoint identity verify all hypotheses of Proposition 20.1. Thus for some $\sigma > 0$ independent of the target,

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_{\text{new}}(\beta_*) - \sigma}.$$

The internal height degrees are allowed to depend on η ; choose $T_1 = Z^{\tau_\eta}$ and then the tail order as in (20.2). The modified average and signal are unchanged by this analysis choice.

Apply Proposition 3.1 with $\sigma_0 = B_{\text{new}}$, $c = -(4+b_{\text{new}})/6$ and, for example, $\omega_0 = \Delta_{\text{new}}/2$. Its hypotheses now all hold: the low bound is at $C_{\text{new}}(B_{\text{new}}) + \omega_0$, the comparison saves σ , and the holomorphic principal factor is within $1/2$ of one. The criterion implies $\beta_* \leq B_{\text{new}}$, contradicting $\Delta_{\text{new}} > 0$. Finally apply the norm-pullback argument of Corollary 14.1 with $B = B_{\text{new}} > 0$. This gives all Dirichlet L -functions, including ζ , and completes the three-part argument. $\square$

APPENDIX A. THETA COEFFICIENT CONVENTIONS

The coefficient formulas, support and normalization are those of [4, equations (5.7)–(5.16) and Appendix A], with the conjugation convention in [1, Proposition 5.1, proof, cusp expansion and infinity coefficient]. Precisely, if d_i^{DR} denotes the coefficient in [4, equation (5.12)], then our three functions are

$$(d_0, d_1, d_2)(\mu) = (\overline{d_1^{\text{DR}}(-\mu)}, \overline{d_{19}^{\text{DR}}(-\mu)}, \overline{d_{10}^{\text{DR}}(-\mu)});$$

the corresponding coefficient formulas are rows 1, 19, 10 of [4, Appendix A], reproducing [3, Table III, p. 151]. The matrices H_j are the representatives $\gamma_1, \gamma_{19}, \gamma_{10}$ specified in [4, Section 5.1, before equation (5.9)]. These locators refer to the version specified in the bibliography.

For clarity, write τ^{DR} for the infinity coefficient in [4, equation (5.7)], and $g^{\text{DR}}(1, c) = \sum_{x \bmod c} (x/c)_3 \check{e}(x/c)$ for its unnormalized cubic Gauss sum. The fourth case of that formula, with its integer exponent equal to zero, gives

$$\tau^{\text{DR}}(\pm \lambda^{-3} c n^3) = 3^{5/2} \frac{|n|}{|c|} \overline{g^{\text{DR}}(1, c)}.$$

[4, equation (5.16), p. 19] uses the conjugated coefficient at $-\mu$, while [4, equation (5.7), p. 17] conjugates the Gauss sum in the infinity coefficient; hence, since $|c| = \sqrt{q_c}$,

$$d_0(\lambda^{-3} c n^3) = \overline{\tau^{\text{DR}}(-\lambda^{-3} c n^3)} = 3^{5/2} |n| \widetilde{g}_3(c).$$

Finally, [4, equations (5.6), (5.15), (5.16)] gives the constant term $3^{5/2} v^{2/3}/2$ for the representative γ_1 , and zero for γ_{19}, γ_{10} , exactly as in Section 4.1.

APPENDIX B. INDEX OF CITED RESULTS

The first table indexes the inputs and deductions in Part I. The second records how the same inputs are used at the new parameters in Parts II and III. Each citation identifies the proved result used in that step.

Reference	Function in the argument
[1, Proposition 2.1]	Holomorphic continuation with a uniform exponent.
[1, Lemmas 4.1–4.4]	Ray characters, Gauss identities and reciprocity phases.
[1, Lemmas 4.5–4.10]	Smooth separation, transform decay and Hecke bounds.
[1, Proposition 5.1; Lemmas 5.2–5.8]	Completed theta functional equation and unweighted mean square.
[1, Lemmas 6.1–6.2; Proposition 6.3]	Balanced additive mean square and $Z^{1/4+\varepsilon}$ estimate.
[1, Lemma 7.1]	Exact scalar Euler quotient and holomorphic correction.
[1, Lemmas 8.1–8.2; Proposition 8.3]	Zero-free rectangles and simultaneous large character polynomials.
[1, Lemma 9.1; Proposition 9.2]	Sextic sieve and preliminary character count.
[1, Lemmas 10.2–10.6]	Contour tails, exponent accounting and principal residues.
[1, Lemma 11.1; Proposition 11.2; Theorem 3.1]	Height choice, first comparison and 11/12 conclusion.
[1, Lemmas 13.1–13.4]	Fixed-ray prime asymptotics and complete correlations.
[1, Corollary 14.1; Lemmas 14.2–14.3]	Divisibility-restricted theta reflection and mean square.
[1, Lemma 15.1; Propositions 15.2–15.3]	Modified completed and additive norms; $Z^{3/16+\varepsilon}$.

Reference	Function in the argument
[1, Proposition 16.1; Lemma 16.2]	Joint local errors and absolute tuple bounds.
[1, Lemmas 17.1–17.3, 17.5–17.6; Corollary 17.4]	Möbius-weighted moment, finite induction and sixth-power amplification.
[1, Lemmas 18.1–18.3]	Unweighted fourth moment and cancellation when the products of scales are equal.
[1, Lemma 19.1; Proposition 19.2]	Prime sum bounds and refined character counts.
[1, Lemmas 20.1–20.2; Proposition 20.3]	Final exponent inequality and uniform parameter choices.
[1, Proposition 11.3]	Transfer to Dirichlet functions.

Input	Use in Parts II and III
[1, Lemma 14.3]	General reflected exponents: Lemmas 16.1 and 22.1.
[1, Proposition 15.2]	Direct bounds with the full additive majorant: Propositions 16.2 and 22.2.
[1, Lemma 7.1]	Extended principal Euler region: Lemma 17.1.
[1, Lemma 13.1]; [1, Lemmas 10.2–10.6]	Normalization, changed-scale residues and tails: Proposition 17.2 and Lemma 19.2.
[1, Proposition 8.3]; [1, Lemmas 17.1, 17.6, 18.1, 19.1]	Fixed- κ count with strict reserves: Proposition 18.1.
[1, Proposition 16.1]; [1, Lemma 16.2]	Joint conductor deficit and changed-scale high exponent: Proposition 18.2.
[1, Lemmas 4.5–4.10, 10.2–10.4, 11.1]	Smooth separation and combination of height estimates: Proposition 20.1.
Proposition 3.1 and Corollary 14.1	Continuation contradiction and Dirichlet transfer at both new boundaries.

REFERENCES

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\operatorname{Re} s > 7/8$, 30 September 2026. Preprint, 30 September version. [Public manuscript](#). All [1] numbering refers to this dated version.
- [2] T. Kubota, *On automorphic functions and the reciprocity law in a number field*, Lectures in Mathematics, Department of Mathematics, Kyoto University, no. 2, Kinokuniya Book-Store, Tokyo, 1969. [Kyoto University repository](#).
- [3] S. J. Patterson, *A cubic analogue of the theta series*, J. Reine Angew. Math. **296** (1977), 125–161. [doi:10.1515/crll.1977.296.125](#).
- [4] A. Dunn and M. Radziwiłł, *Bias in cubic Gauss sums: Patterson’s conjecture*, Ann. of Math. (2) **200** (2024), no. 3, 967–1057. [doi:10.4007/annals.2024.200.3.3](#). Equation and appendix locators in this paper refer to [arXiv:2109.07463v3](#), 14 May 2024.
- [5] L. Goldmakher and B. Louvel, *A quadratic large sieve inequality over number fields*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 2, 193–212. [doi:10.1017/S0305004112000370](#).
- [6] M. N. Huxley, *The large sieve inequality for algebraic number fields*, Mathematika **15** (1968), no. 2, 178–187. [doi:10.1112/S0025579300002540](#).
- [7] D. R. Heath-Brown, *Kummer’s conjecture for cubic Gauss sums*, Israel J. Math. **120**, part A (2000), 97–124. [doi:10.1007/s11856-000-1273-y](#).
- [8] J. Maynard and K. Pratt, *Half-isolated zeros and zero-density estimates*, Int. Math. Res. Not. IMRN **2024**, no. 19, 12978–13014. [doi:10.1093/imrn/rnae191](#). The appendix locator refers to the [arXiv manuscript](#).
- [9] V. Blomer, L. Goldmakher, and B. Louvel, *L-functions with n -th-order twists*, Int. Math. Res. Not. IMRN **2014**, no. 7, 1925–1955. [doi:10.1093/imrn/rns257](#).

- [10] J. Thorner and A. Zaman, *A unified and improved Chebotarev density theorem*, Algebra Number Theory **13** (2019), no. 5, 1039–1068. [doi:10.2140/ant.2019.13.1039](https://doi.org/10.2140/ant.2019.13.1039).
- [11] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, Ann. of Math. (2) **203** (2026), no. 2, 623–675. [doi:10.4007/annals.2026.203.2.6](https://doi.org/10.4007/annals.2026.203.2.6). The theorem locator refers to [arXiv:2405.20552v2](https://arxiv.org/abs/2405.20552v2), 7 April 2026.
- [12] OpenAI, *The Quasi-Riemann Hypothesis (alternate 11/12 proof)*, preprint, 5 October 2026. [Public manuscript](#).
- [13] OpenAI, *The quasi-Riemann hypothesis: scope of the Lean formalization*, repository documentation, accessed 7 October 2026. [Scope and comparator links](#).
- [14] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004. [doi:10.1090/coll/053](https://doi.org/10.1090/coll/053).
- [15] B. Liu, *Lean formalizations of the rational and algebraic zero-free extensions*, accompanying source-code repository, 2026, Lean 4.34.1. [GitHub repository for the paper and Lean formalizations](#). Paths below are relative to the repository root. The manuscript correspondence entry is `formalization/RHZeroFreeExtension/CombinedPaperVerification.lean`, namespace `CombinedPaper`. The complete rational entry is `formalization/RHZeroFreeExtension.lean`, namespace `RHZeroFreeExtension`; the complete algebraic entry is `formalization/RHZeroFreeExtension/V2FullVerification.lean`, namespace `RHV2`. In the latter two namespaces the public conclusions are `beta_star_le`, `hecke_nonzero`, `dirichlet_nonzero` and `zeta_nonzero`. The verifier `formalization/verify.py` builds the complete import closure and replays the 22 correspondence theorems with `-trust=0`; the reported transitive axioms are `propext`, `Classical.choice` and `Quot.sound`.
- [16] S. Bernstein, Démonstration du théorème de Weierstrass fondée sur le calcul des probabilités, *Communications de la Société mathématique de Kharkow*, 2nd series, **13** (1912), no. 1, 1–2. [Original paper and bibliographic record](#).
- [17] R. T. Farouki, The Bernstein polynomial basis: A centennial retrospective, *Computer Aided Geometric Design* **29** (2012), no. 6, 379–419. [doi:10.1016/j.cagd.2012.03.001](https://doi.org/10.1016/j.cagd.2012.03.001).

DEPARTMENT OF MATHEMATICS, PURDUE UNIVERSITY, WEST LAFAYETTE, IN, 47907, USA
 Email address: liu2053@purdue.edu